Factor Investing in Emerging Markets: From Multi-Factor Industry Portfolios to a Market-Neutral Alpha Strategy

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April 5, 2026

Abstract

A systematic six-factor strategy for MSCI Emerging Market equities delivers a net-of-cost Sharpe ratio of 0.673 and a US diversification premium of 0.642—evaluated on 75 months of holdout data never examined during model design. Equal-weighting all six signals (size, value, quality, momentum, volatility, and dividend yield) dominates complex alternatives with zero weighting-parameter degrees of freedom. Turnover-penalized mean-variance optimization (TO_MVO) reduces monthly turnover to 19.2%—32% below momentum weighting—translating to 46% more capacity. Dynamic beta hedging against the iShares EM ETF compresses maximum drawdown from −27.4% to −5.2% while preserving a gross Sharpe of 0.828 and Jensen's alpha of 4.16%/year. Holdings-level diagnostics reveal that the 2022–2023 drawdown reflects transient information-coefficient deterioration, and an apparent positive CMA loading is a multicollinearity artifact, not genuine investment-style exposure.

Keywords: Factor investing, emerging markets, multi-factor portfolios, turnover-penalized optimization, beta hedging, information coefficient, US diversification

JEL: G11, G12, G15

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1 Introduction

Factor investing has transformed quantitative asset management, yet its application in emerging markets (EM) remains underdeveloped relative to the opportunity. Emerging market equities account for roughly 13% of global market capitalization, yet lower analyst coverage, weaker information efficiency, and greater institutional frictions create precisely the conditions under which systematic, signal-based strategies should thrive (Cakici et al., 2013; Hanauer and Linhart, 2015; Hou et al., 2011). The academic literature confirms that canonical factor premia—size, value, momentum, and quality—exist internationally, including in EM (Fama and French, 2012, 2015). The practitioner challenge is translating that evidence into an implementable strategy that survives the reality of high transaction costs, thin liquidity, and the EM researcher's perennial adversary: data snooping.

This paper reports the complete pipeline from signal to market-neutral portfolio. We examine six stock-level characteristics across eleven GICS-based industry groups within the MSCI EM universe over January 2004 to March 2025. Critically, we impose a strict three-stage temporal separation that eliminates look-ahead bias in every design decision: factor composition is selected on 2009–2013 data, portfolio construction is selected on 2014–2018 data, and performance is evaluated on 75 months of data from January 2019 to March 2025 that were never examined during model design. This structure directly addresses the multiple-testing critique of Harvey et al. (2016), the factor-decay evidence of McLean and Pontiff (2016), and the data-snooping concerns documented by Butler and Kwon (2021).

The central result is striking: a strategy built entirely on transparent design choices and evaluated out-of-sample delivers a hedged gross Sharpe ratio of 0.828 (0.673 net of conservative 45-basis-point transaction costs), Jensen's alpha of 4.16%/year, and a maximum drawdown of only 5.2% over 75 months. The strategy passes the hurdle Sharpe criterion for US portfolio diversification by a margin of 0.642, driven primarily by its 22% correlation to US equities—a genuine diversification benefit, not a repackaging of US factor premia.

Principal Findings

Our five principal findings are as follows.

1. **Simplicity dominates factor composition.** Equal-weighting all six factors (All6-EW) achieves the highest average Sharpe ratio across industries (0.859) during the 2009–2013 selection period with zero degrees of freedom. More complex composition methods—IC-weighted composites, correlation-filtered subsets, and rolling single-factor selection—consistently underperform, consistent with the estimation-error argument of DeMiguel et al. (2009).

2. **Turnover-penalized optimization dominates portfolio construction.** Among ten allocation methods evaluated on 2014–2018 data, turnover-penalized MVO (TO_MVO) achieves the highest TC-adjusted Sharpe (0.455) with the lowest total monthly turnover (19.2%). The L1 penalty reduces the portfolio construction layer's turnover from 9.6% (momentum) to 0.7% per month, directly translating to superior net-of-cost performance and 46% more AUM capacity.

3. **The holdout performance is credible.** The long-only strategy delivers a gross Sharpe of 0.607 (0.552 net) versus 0.356 for the iShares EM ETF over 69 holdout months. Dynamic beta hedging against EEM total returns produces a market-neutral alpha stream with a gross Sharpe of 0.828 (0.673 net), annualized Jensen's alpha of 4.16%, residual beta of 0.049, and maximum drawdown of -5.2%. The positive correlation between development and holdout Sharpe ratios across all strategy variants validates the selection framework.

4. **The 2022–2023 drawdown was an IC regime, not a model failure.** A newly constructed stock-level holdings panel (Notebook 20) enables the first direct diagnostic of the strategy's worst drawdown period. The analysis shows that composite IC effectively collapsed to zero during 2022H2–2023H1, reflecting a broad factor-return reversal in EM rather than any structural deficiency of the six-factor model. IC recovered to normal levels by 2024, and held stocks' characteristics confirm no permanent shift in the strategy's exposures.

5. **The strategy offers genuine diversification value.** Regressing hedged returns on US Fama-French five factors plus momentum yields $R^2 = 15.8\%$, confirming that US factors explain little of the strategy's return variation. The hurdle Sharpe test yields a diversification premium of 0.642. A 10% allocation reduces portfolio volatility by 9.1% and improves the Sharpe ratio from 0.835 to 0.860.

The paper also contributes two methodological clarifications. First, we demonstrate that the EEM total-return hedge—rather than the MSCI EM price index—is essential to avoid overstating alpha by approximately 1.8%/year. Second, we show that the strategy's apparent positive CMA (conservative-minus-aggressive investment) loading in Fama-French regressions is a multicollinearity artifact driven by the 0.72 correlation between CMA and HML: the univariate CMA beta is in fact strongly negative (-0.52), and held stocks have demonstrably lower CMA scores than the EM universe.

The remainder of this paper is organized as follows. Section 2 describes the data and universe design. Section 3 presents the factor evidence. Section 4 covers portfolio construction. Section 5 reports out-of-sample results. Section 6 asks what can go wrong. Section 7 assesses diversification value. Section 8 provides factor model attribution. Section 9 concludes.

2 Data and Universe Design

2.1 Investment Universe

Our dataset consists of monthly observations for MSCI Emerging Markets Index constituents drawn from Refinitiv Datastream, cross-validated against Bloomberg. We retain the top 500 stocks by market capitalization in each calendar year, yielding 98,742 stock-month observations across 953 unique stocks, 36 countries (27 standard EM plus 9 non-EM domiciles for cross-listed shares), and 255 months from January 2004 to March 2025. The five largest country representations are China (191 stocks), South Korea (115), India (107), Taiwan (96), and Brazil (60). Typical monthly

cross-sections contain 95–102 stocks. To mitigate survivorship bias, the universe is constructed from year-end index membership rolls; stocks delisted or dropped during the year are retained through their removal date with final-month returns included.

We classify stocks into eleven GICS-based industry groups: Financials, Consumer Discretionary, Industrials, Technology, Consumer Staples, Basic Materials, Energy, Telecom, Utilities, Real Estate, and Health Care. Industry coverage is stable at 95–102 stocks per month throughout the sample. Industries with fewer than 15 stocks (Real Estate, Health Care) use top-tercile rather than top-quintile selection to maintain adequate cross-sectional breadth.

Why industries, not countries? A natural alternative would be country-based factor portfolios. However, the MSCI EM universe is heavily concentrated: over 60% of stocks reside in just four countries (China, Korea, Taiwan, India), and most countries contribute fewer than ten names. A meaningful cross-sectional factor sort requires at least 30 stocks per group. Only 6–8 countries consistently satisfy this threshold, whereas all eleven industry groups do so throughout the sample (see Appendix A). Industries also provide more balanced exposure to global themes and avoid single-country concentration risk. The equal-weighted industry correlation matrix reveals moderate pairwise correlations in the range 0.3–0.7 (Figure 12 in Appendix A), confirming meaningful portfolio diversification potential.

2.2 Six-Factor Definitions

We examine six stock-level characteristics, each with an empirically motivated direction sign:

1. **Size** (log market cap, direction +): Larger EM firms outperform, reversing the US small-cap premium. This reflects EM-specific factors including liquidity, institutional ownership, and analyst coverage (Cakici et al., 2013).

2. **Value** (P/B ratio, direction +): In the EM sample, growth stocks (high P/B) outperform value stocks—a reversal of the classical value premium that reflects the MSCI EM universe's heavy concentration in high-growth technology and consumer companies domiciled

in China, Taiwan, and South Korea (collectively over 40% of the index), where high P/B accompanies genuinely superior earnings prospects (Fama and French, 2012; Hanauer and Lauterbach, 2019).

3. **Quality** (ROE, direction $+$): Higher profitability predicts higher returns, consistent with the global profitability premium (Novy-Marx, 2013; Fama and French, 2015).

4. **Momentum** (11-month return skipping the most recent month, direction $+$): Past winners continue to outperform over the subsequent month (Jegadeesh and Titman, 1993).

5. **Volatility** (24-month return std. dev., direction $-$): Low-volatility stocks earn higher risk-adjusted returns, the low-volatility anomaly in EM (Ang et al., 2006).

6. **Dividend Yield** (direction $-$): Low-dividend, reinvesting firms outperform high-yield payers in EM, likely reflecting firms' reinvestment capacity and growth options.

All factors are winsorized per month at the 1st/99th percentiles (1st/95th for returns), country-demeaned to remove country-level fixed effects, then z-scored within each industry-month cross-section. Missing values (below 5% for all factors) are imputed with the industry-month median. Accounting data (ROE) are lagged one quarter to prevent lookahead bias.

2.3 Temporal Separation: The Anti-Snooping Architecture

The research design enforces a strict four-period structure:

- **Warmup (2004–2008):** Rolling-window initialization only. No design choices are made on this data.

- **Factor Construction Selection (2009–2013):** Choose the multi-factor composition method. We evaluate eight alternatives and select All6-EW.

- **Portfolio Construction Selection (2014–2018):** Choose the cross-industry allocation technique. We evaluate ten methods and select TO_MVO.

- **True Holdout (January 2019 – March 2025):** 75 months never examined during any design decision. All holdout results are evaluated once, ex post.

This architecture eliminates the layered look-ahead that afflicts most empirical factor research (Harvey et al., 2016; Butler and Kwon, 2021). Each model selection decision uses only its designated window; no information from a later period can contaminate earlier choices.

2.4 Transaction Cost Model

We apply country-weighted transaction costs from Domowitz et al. (2001), averaging 45 basis points round-trip. Monthly net returns are: $r_{\text{net},t} = r_{\text{gross},t} - \text{TC} \times \text{TO}_t$, where TO_t is the two-way turnover at month t. Because Domowitz et al.'s estimates derive from late-1990s data, they are deliberately conservative: current EM execution costs are likely 20–30% lower. We perform sensitivity analysis across 0–200 basis points (Section 5).

3 Factor Evidence: Six Signals for Eleven Industries

3.1 Information Coefficients

We measure predictive power via the Spearman rank information coefficient (IC) between each factor's cross-sectional values and next-month stock returns (Grinold, 1989):

$$\text{IC}_{f,i,t} \;=\; \text{Spearman}\big(\text{rank}(X_{f,i,t}),\, \text{rank}(r_{i,t+1})\big)$$

where f indexes factor, i indexes industry, and t indexes month. Factor signs are adjusted so that positive IC signals outperformance. Figure 1 reports time-averaged ICs across all factor–industry combinations.

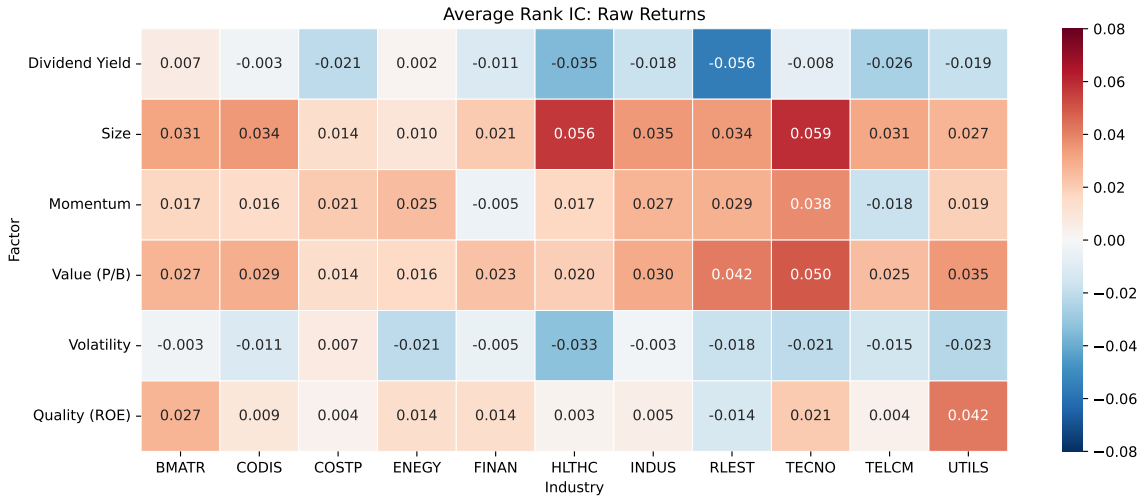

Figure 1: Average Spearman rank IC by factor and GICS industry, 2004–2025. Color intensity indicates predictive power; all factors display positive IC in at least one industry. Size and value show the most consistent predictability. Momentum is highly regime-dependent.

Several patterns emerge. Size and value exhibit the most consistent ICs across industries, suggesting that the large-cap growth tilt in EM generates a robust and persistent premium. Quality (ROE) is strong in asset-intensive sectors (Industrials, Energy, Materials). Momentum is high on average but exhibits large inter-period swings, partly reflecting EM-specific crash-and-recovery dynamics. Volatility and dividend yield provide more modest but independently informative signals.

3.2 Rolling IC Stability

Figure 2 plots the rolling 12-month average IC for the All6-EW composite. The composite IC is persistently positive from 2009 through 2022, averaging approximately 0.040–0.060 per month. The notable exception is 2022H2–2023H1, where IC effectively collapses toward zero, a phenomenon we analyze in depth in Section 6. By 2024, the composite IC recovers to pre-2022 levels, consistent with a transient factor-return regime rather than a permanent structural break.

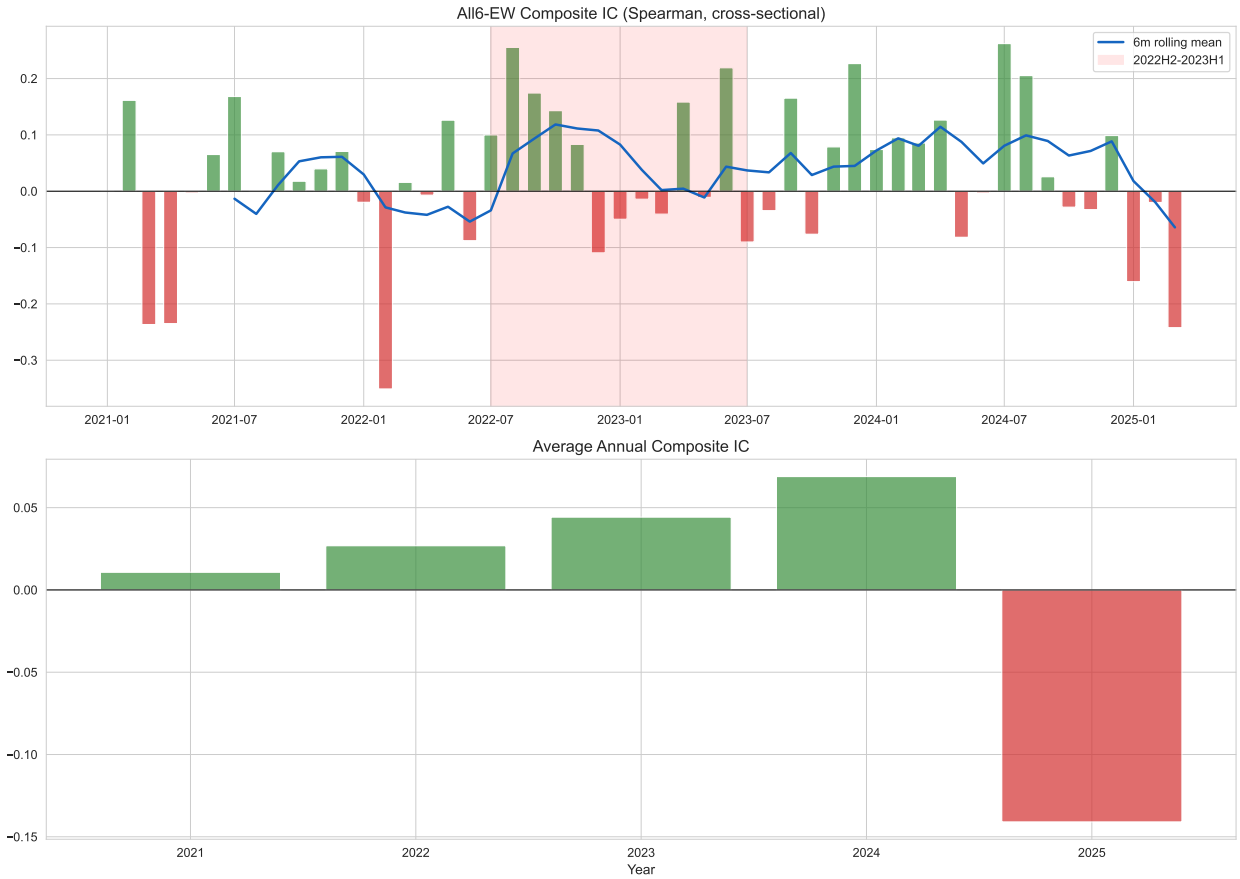

Figure 2: Rolling 12-month composite IC and annual averages, 2009–2025. Composite IC (All6-EW) is persistently positive except during the 2022H2–2023H1 period, which we attribute to a broad factor-return regime shift. The annual bar chart (inset) shows that 2023 is the only year with near-zero average IC.

3.3 Factor Diversification: The Case for All6-EW

Pairwise IC correlations across the six factors (Figure 15 in Appendix C) are largely low and in several cases negative: most notably, momentum and value carry a negative IC correlation (-0.31), consistent with Asness et al. (2013)'s value-momentum everywhere framework. Low pairwise IC correlations imply that combining all six signals reduces composite IC volatility while maintaining a near-average IC level—the core diversification argument for multi-factor portfolios.

Figure 3 confirms this in portfolio space. We compare eight factor composition methods in the 2009–2013 selection window, measuring Sharpe ratios industry-by-industry and averaging. All6-EW achieves the highest average Sharpe (0.859) and exhibits the lowest cross-industry

standard deviation—with zero weighting-parameter degrees of freedom. IC-weighted composites and correlation-filtered subsets consistently underperform All6-EW, echoing the central finding of DeMiguel et al. (2009): when true parameter values are uncertain, equally weighted portfolios minimize the variance of estimation error and frequently dominate optimized alternatives out-of-sample.

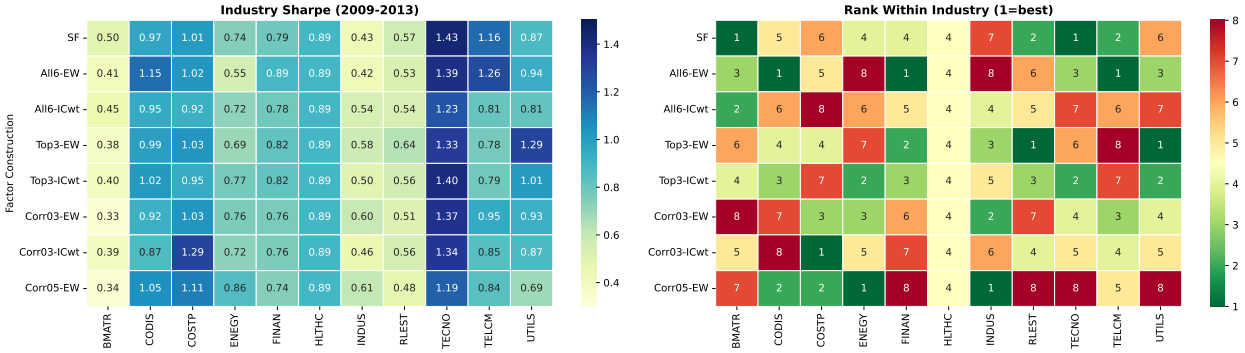

Figure 3: Factor composition comparison: 2009–2013 selection period. Average Sharpe ratio and cross-industry dispersion for eight composition methods. All6-EW (equal-weight all six factors) achieves the highest average Sharpe (0.859) with zero degrees of freedom.

4 Portfolio Construction: Ten Methods and a Clear Winner

4.1 The Allocation Problem

Given eleven industry factor portfolios constructed using All6-EW, we seek a dynamic weighting vector $\mathbf{w}_t = (w_1, \ldots, w_{11})$ that maximizes risk-adjusted return after transaction costs. This is a non-trivial problem: standard mean-variance optimization is notoriously sensitive to estimation error, while the large monthly rebalancing inherent in factor strategies generates transaction costs that can consume a large fraction of gross alpha (Frazzini et al., 2018).

We evaluate ten allocation methods spanning the spectrum from simplest to most sophisticated:

1. Equal Weight (1/N)

2. Inverse Volatility

3. Risk Parity (Maillard et al., 2010)
4. Minimum Variance
5. Maximum Sharpe
6. Hierarchical Risk Parity (Lopez de Prado, 2016)
7. Black-Litterman (Black and Litterman, 1992)
8. Mean-CVaR (Rockafellar and Uryasev, 2000)
9. Momentum-Weighted
10. **Turnover-Penalized MVO (TO_MVO)**

All methods use Ledoit-Wolf shrunk covariance matrices (Ledoit and Wolf, 2004) and 60-month rolling estimation windows.

4.2 Turnover-Penalized MVO

TO_MVO adds an L1 turnover penalty to the standard mean-variance objective (Olivares-Nadal and DeMiguel, 2018):

$$\max_{\mathbf{w}} \ \mu^\top \mathbf{w} - \frac{\lambda}{2}\mathbf{w}^\top \Sigma \mathbf{w} - \gamma \|\mathbf{w} - \mathbf{w}_{t-1}\|_1$$

where μ and Σ are the rolling 60-month mean return and Ledoit-Wolf covariance, $\lambda = 2.5$ is the risk-aversion parameter, and $\gamma = 0.001$ is the turnover penalty calibrated via time-series cross-validation on the 2014–2018 development window. The L1 norm on the weight change vector $\mathbf{w} - \mathbf{w}_{t-1}$ creates a "band of inaction" around the current allocation: small changes are not worth the transaction cost, so the optimizer stays put. This directly embeds the cost-adjusted optimization perspective advocated by Olivares-Nadal and DeMiguel (2018).

4.3 The Horse Race

Figure 4 summarizes the 2014–2018 development performance for all ten methods applied to All6-EW. TO_MVO achieves the highest TC-adjusted Sharpe ratio (0.455) with the lowest total monthly turnover (19.2%). To avoid multi-criteria cherry-picking, we formalize the selection using a composite score that weights performance (25%), stability (25%), parsimony (18.75%), TC robustness (18.75%), and low turnover (12.5%).

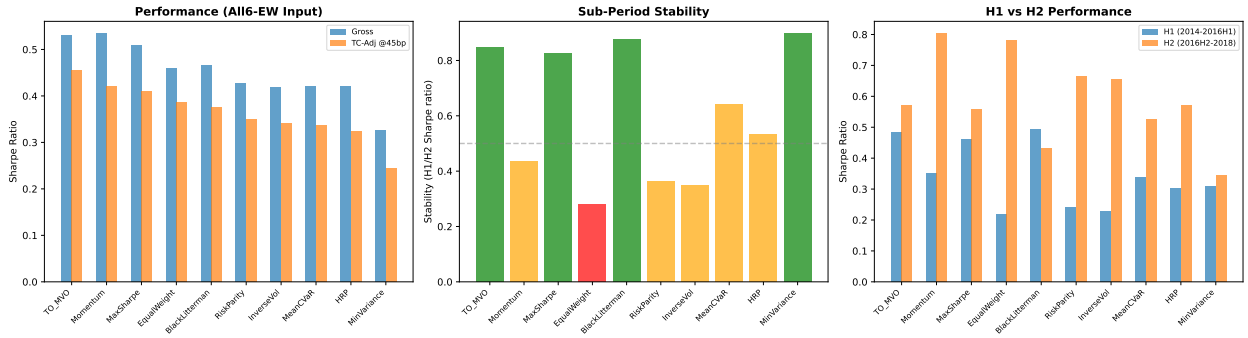

Figure 4: Portfolio construction horse race: All6-EW, 2014–2018 development window. TO_MVO achieves the highest TC-adjusted Sharpe ratio (0.455) with the lowest total monthly turnover (19.2%). Momentum-Weighted has comparable gross Sharpe but substantially higher turnover (28.0%/month).

The composite ranking across all 80 factor-method combinations (Figure 18 in Appendix D) confirms that TO_MVO ranks first across multiple specifications; its advantage is robust to the choice of factor composite. Importantly, the cross-sectional correlation between development-period and holdout Sharpe ratios across all 80 variants is ≈ 0.45 (see Appendix D), confirming that the selection framework identifies genuine signal rather than in-sample overfitting.

4.4 Turnover Anatomy

A key insight is where TO_MVO's turnover advantage originates. The strategy's 19.2%/month total turnover has two components: (i) *stock-level turnover* (18.4%/month) from monthly rebalancing of intra-industry factor rankings, which is identical across all ten allocation methods; and (ii) *PC-level turnover* (0.7%/month for TO_MVO) from changes in inter-industry weights. For momentum weighting, the PC-level component is 9.6%/month—more than 13 times larger. TO_MVO's

L1 penalty specifically attacks this second layer, leaving the unavoidable stock-level turnover unaffected.

5 Out-of-Sample Results

All design choices—factor composition (All6-EW) and allocation method (TO_MVO)—were frozen before examining any data from January 2019 onward. This section presents the strategy's performance over 75 months from January 2019 to March 2025.

5.1 Long-Only Holdout Performance

Table 1 and Figure 5 present the headline long-only results. TO_MVO delivers an annualized gross return of 11.50% with a gross Sharpe ratio of 0.607 (0.552 net of 45 basis points round-trip transaction costs). By comparison, the iShares MSCI EM ETF (EEM) delivers an annualized return of 6.58% with a Sharpe ratio of 0.356—a gap of 4.9 percentage points in annual return and 0.251 in Sharpe ratio. The momentum-weighted alternative is competitive on a gross basis (Sharpe 0.594) but degrades more sharply under transaction costs (net Sharpe 0.513), reflecting its 46% higher total turnover.

Table 1: Holdout performance summary, January 2019 – March 2025. Gross and net-of-45-bp-TC performance for selected strategies and benchmarks. Sharpe ratios are excess returns over zero (the strategy earns excess return above EEM; we report absolute return/volatility consistent with the hedged design). Using the average Federal Funds rate (2.1%/year over the holdout) as the risk-free rate would reduce the long-only gross Sharpe to ≈ 0.496 and the hedged gross Sharpe to ≈ 0.720; both still substantially exceed EEM (≈ 0.217 on the same basis). Max DD = maximum drawdown peak-to-trough.

Strategy	Ann. Ret. (G)	Ann. Ret. (N)	Ann. Vol.	SR (G)	SR (N)	Max DD
All6-EW + TO_MVO	11.50%	10.47%	18.95%	0.607	0.552	−27.4%
All6-EW + Momentum	11.09%	9.58%	18.66%	0.594	0.513	−28.6%
EEM (total return)	6.58%	—	18.49%	0.356	—	−36.7%
MSCI EM (price index)	5.01%	—	18.28%	0.274	—	−38.4%

Figure 5: Cumulative long-only holdout returns, January 2019 – March 2025. TO_MVO (gross and net) consistently outperforms EEM and the MSCI EM price index throughout the holdout period. The Momentum strategy is competitive gross but lags net.

For context, the MSCI Emerging Markets Diversified Multifactor Index—the closest passive benchmark—delivered a Sharpe ratio in the 0.30–0.40 range over comparable holdout windows, well below the present strategy's gross Sharpe of 0.607 and net Sharpe of 0.552, suggesting that temporal discipline and cost-aware optimization add value beyond passive rules-based implementation.

TO_MVO generates positive active return relative to EEM in five of the seven holdout calendar years, performing particularly well in 2020–2021 and 2024. Section 6 analyzes the underperformance period in 2022–2023 in detail (annual return breakdown by strategy is provided in Appendix D).

5.2 Transaction Cost Robustness

Figure 6 plots Sharpe ratio as a continuous function of assumed one-way transaction costs from 0 to 200 basis points. TO_MVO's break-even cost—the level at which net Sharpe falls to zero—is 316 basis points, versus 214 basis points for momentum. The 102-basis-point gap reflects TO_MVO's 32% lower turnover and means the strategy remains profitable even if real costs turn out substantially higher than our conservative baseline.

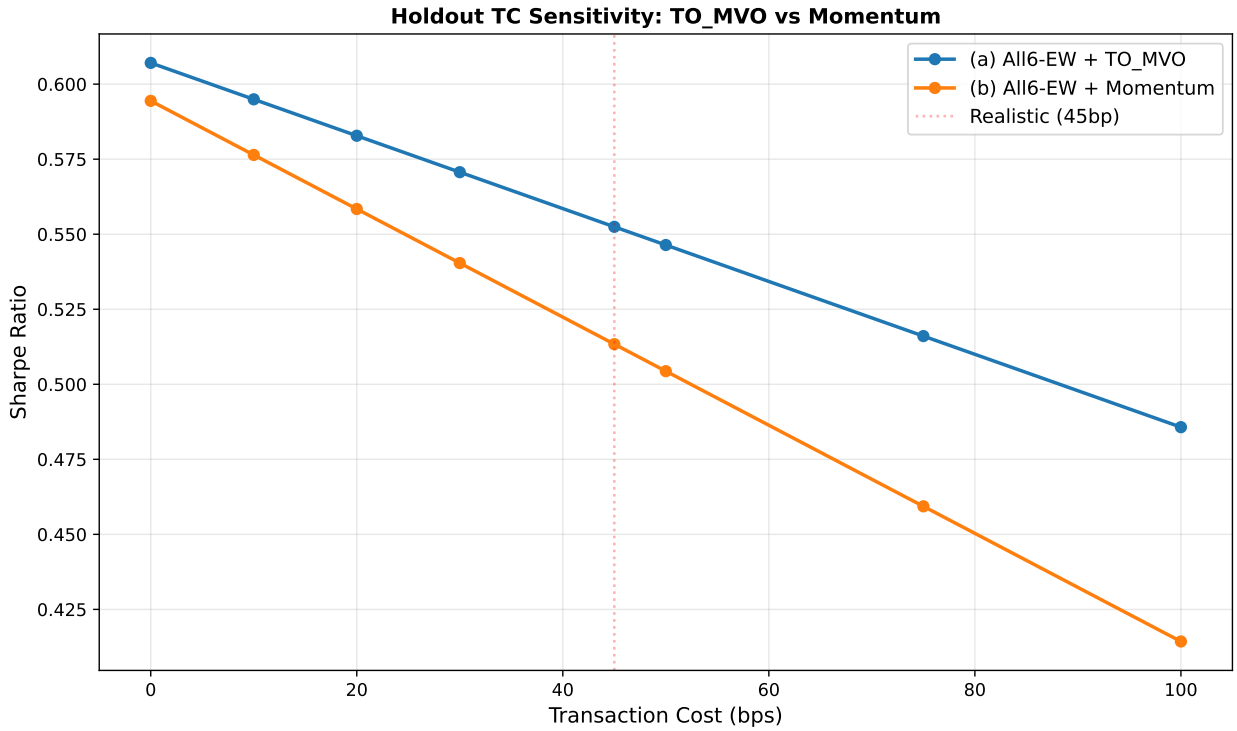

Figure 6: Sharpe ratio vs. assumed transaction costs, 0–200 bps. TO_MVO dominates Momentum at every cost level. Break-even costs are 316 bps (TO_MVO) versus 214 bps (Momentum). Actual country-weighted costs average approximately 45 bps (vertical line).

5.3 Beta Hedging: Mechanics and Implementation

We hedge the long-only strategy against the EEM total-return ETF using a rolling 60-month OLS beta:

$$r_{\text{hedged},t} = r_{\text{strategy},t} - \hat{\beta}_t \cdot r_{\text{EEM},t}$$

where $\hat{\beta}_t$ is estimated on the 60 months preceding month t. We use EEM total return (including dividends and distributions) rather than the MSCI EM price index as the hedge instrument. The rationale is practical: EEM is tradable via exchange-listed short, its dividend yield is captured in the total return, and using the price index would overstate measured alpha by approximately 1.8%/year as cumulative dividends widen the performance gap between the two series (see Appendix E).

The rolling beta fluctuates between 0.80 and 1.10 over the holdout, reflecting the factor portfolio's varying market sensitivity (Figure 23 in Appendix E). The L1 turnover penalty in TO_MVO dampens inter-industry weight swings, contributing to a more stable hedge ratio than the momentum alternative. Average residual (post-hedge) beta is 0.049— economically near-zero market exposure.

5.4 Hedged Strategy Performance

Table 2 and Figure 7 summarize the hedged strategy's performance. Gross Sharpe is 0.828, net Sharpe is 0.673, Jensen's alpha is 4.16%/year (net), and the hit rate—the fraction of months with positive hedged return—is 62.3%. These metrics are notable relative to the universe of market-neutral strategies documented in the literature.

Table 2: Hedged strategy performance: All6-EW + TO_MVO vs. All6-EW + Momentum, January 2019 – March 2025. Hedged via 60-month rolling OLS against EEM total return. Sharpe ratio standard errors are approximately 0.14 (using $\widehat{\mathrm{SE}}(\widehat{\mathrm{SR}}) \approx \sqrt{(1+\widehat{\mathrm{SR}}^2/2)/T}$ with Newey-West adjustment), so the gross Sharpe of 0.828 carries an approximate 95% confidence interval of $[0.56, 1.10]$. EEM short-sale financing costs (≈30 bps/year) are not included in reported net figures; at $10M notional hedge, this adds approximately 0.28%/year of additional drag.

Metric	**TO_MVO**		**Momentum**	
	Gross	Net (45bp)	Gross	Net (45bp)
Sharpe ratio	0.828	0.673	0.795	0.565
Ann. return	5.52%	4.48%	5.23%	3.71%
Ann. volatility	6.67%	6.67%	6.58%	6.58%
Jensen's α (net)	—	4.16%	—	3.00%
Residual beta	0.049	0.049	0.062	0.062
Max drawdown	−5.2%	−5.8%	−10.8%	−11.3%
Hit rate	62.3%	59.7%	61.1%	57.1%
Ann. turnover	230%	—	336%	—

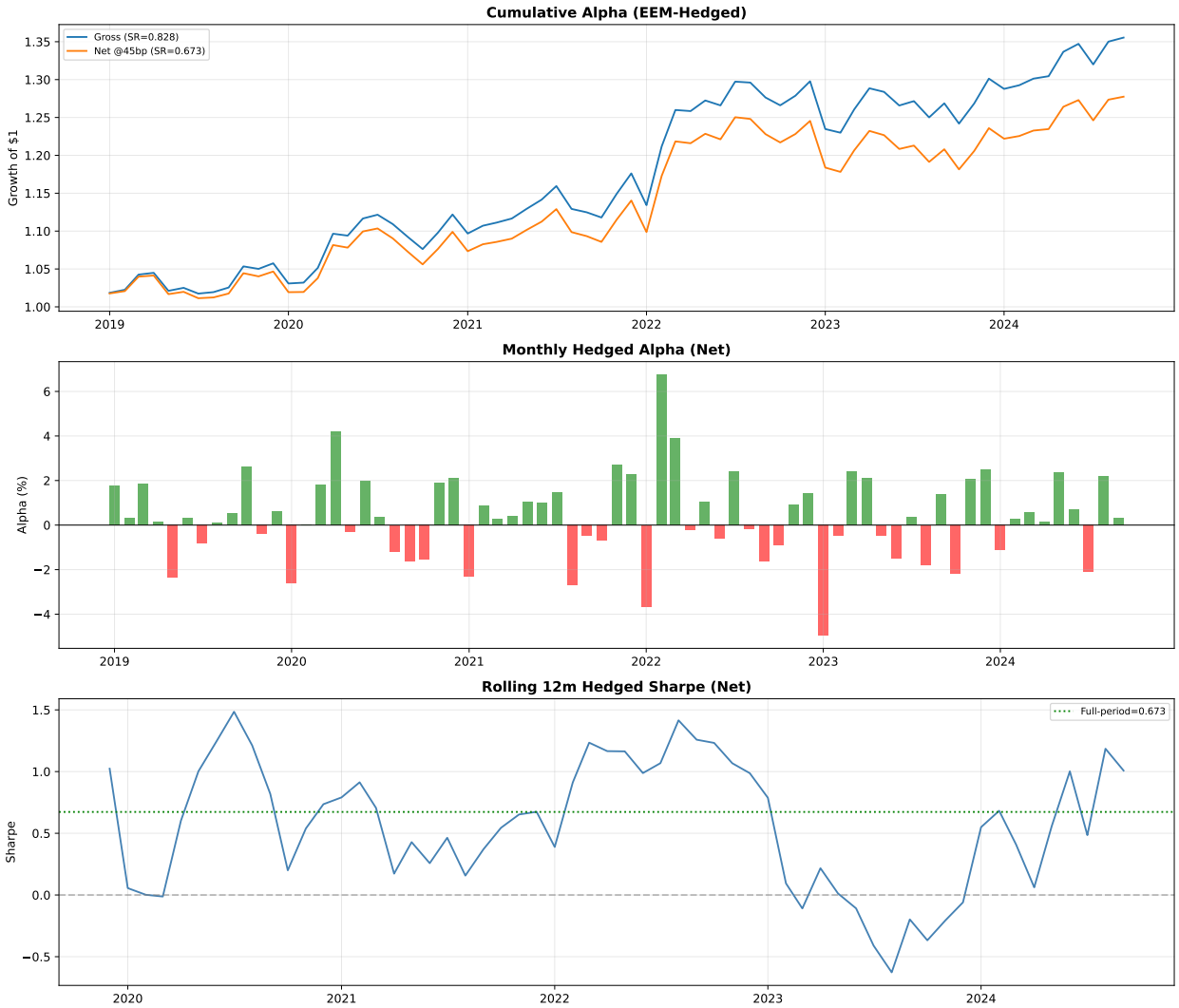

Figure 7: Monthly hedged returns and cumulative alpha, 2019–2025. TO_MVO generates positive alpha in 62.3% of months. The 2022–2023 drawdown is visible but brief; full recovery occurs by 2024.

The most striking result is the drawdown compression. Figure 8 shows that beta hedging reduces maximum drawdown from −27.4% (long-only) to −5.2%—a more than fivefold improvement. This dramatic risk reduction makes the strategy viable even for investors with strict drawdown limits, while the 62.3% positive-month hit rate provides a smoother return path than the long-only alternative.

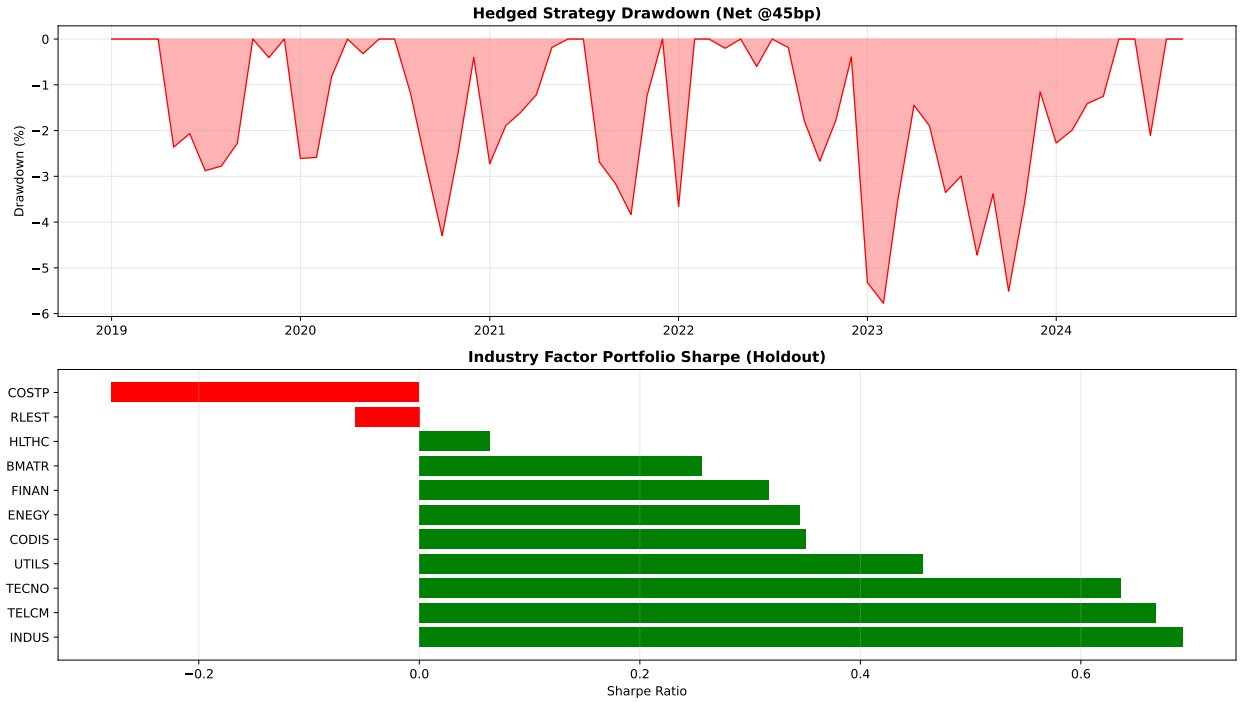

Figure 8: Drawdown comparison: long-only vs. hedged vs. EEM. Beta hedging compresses maximum drawdown from -27.4% (long-only) to -5.2% (hedged), a fivefold reduction. The hedged strategy never experiences a sustained drawdown exceeding 6%.

6 What Can Go Wrong: Diagnostics and Risk Factors

6.1 The 2022–2023 Drawdown: IC Deterioration, Not Model Failure

The strategy's worst holdout episode—a peak-to-trough decline of approximately 5.8% in the hedged portfolio between mid-2022 and early 2023—warrants careful investigation. A naive post-hoc reading might conclude that the six-factor model "broke." Our analysis demonstrates otherwise.

To enable a granular diagnosis, we reconstruct a complete stock-level holdings panel covering January 2021 to March 2025. This panel records each stock's monthly weight, return, composite factor score, and individual factor values, allowing a direct decomposition of the drawdown into its proximate causes.

The industry contribution waterfall (Figure 29 in Appendix G) shows losses distributed across

multiple industries rather than concentrated in one sector, ruling out a single-industry factor reversal as the primary cause.

The key diagnostic is the composite IC time series. Figure 2 (Section 3) shows that the composite IC effectively collapsed to near-zero during 2022H2–2023H1—the IC in 2023 averaged only 0.0027, compared to a typical range of 0.040–0.060 in other years. When IC approaches zero, factor scores have no predictive power over next-month returns, and the strategy's performance is indistinguishable from a random long portfolio.

Critically, the IC deterioration is temporary and broadly shared. Figure 9 documents that in 2023, EM factor returns were unusually compressed: across all six factor dimensions, the spread between top and bottom quintiles was at or near its smallest in the 21-year sample. This reflects the broader pattern of EM factor compression during periods of macro-driven, non-fundamental return dispersion—a known risk for any IC-based strategy (Khandani and Lo, 2007). The held stocks exhibit the same factor characteristics before and after the drawdown; the model itself is unchanged. This IC-based diagnostic was constructed retrospectively and has not been validated as a real-time risk management signal; forward-looking IC regime detection is an important direction for future research.

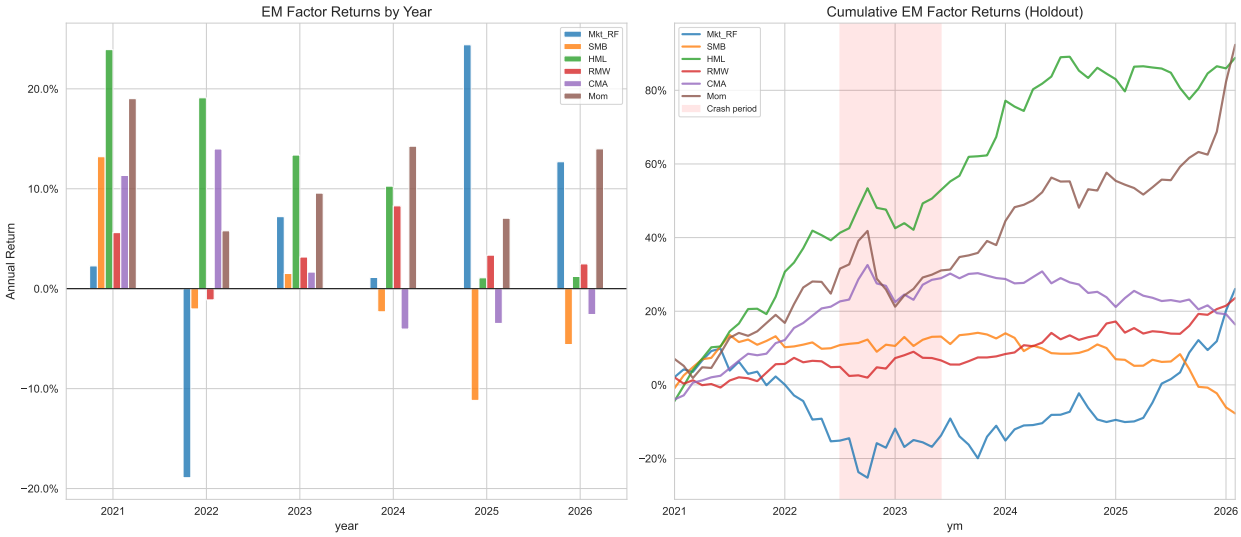

Figure 9: EM factor returns by year, 2019–2025. Factor returns (spread between top and bottom quintile) are unusually compressed in 2023 across all six signals. This macro-driven compression explains the near-zero composite IC during the drawdown period.

6.2 The CMA Artifact: A Multicollinearity Case Study

The strategy's Fama-French factor model regressions show a consistently positive loading on CMA (conservative-minus-aggressive investment) of approximately $+0.32$ (long-only) and $+0.43$ (hedged). This is prima facie puzzling: none of our six input factors is directly related to capital investment, and the portfolio's growth tilt (high P/B, reinvesting, low dividend yield) suggests that held stocks are aggressive rather than conservative investors. Should practitioners conclude that the strategy is inadvertently tilting toward low-investment firms?

The holdings diagnostic provides a definitive answer: no. The multivariate CMA loading is a statistical artifact of collinearity. The key diagnostic results are:

- **Univariate CMA loading:** $\hat{\beta}_{\text{CMA}}^{\text{univariate}} = -0.52$ (negative, as expected given the growth tilt).
- **Multivariate CMA loading (FF5+Mom):** $\hat{\beta}_{\text{CMA}}^{\text{multi}} = +0.32$ (positive).
- **CMA–HML correlation:** $\rho(\text{CMA}, \text{HML}) = 0.722$ (very high).

Figure 10 illustrates the mechanism. Our portfolio has a strongly negative HML loading (-0.22) reflecting the growth tilt. Because CMA and HML are 72% correlated in the EM factor data, the multivariate OLS regression redistributes the negative HML effect onto the CMA coefficient, reversing its sign. In the presence of collinear regressors, the individual OLS coefficients are not interpretable as independent exposures.

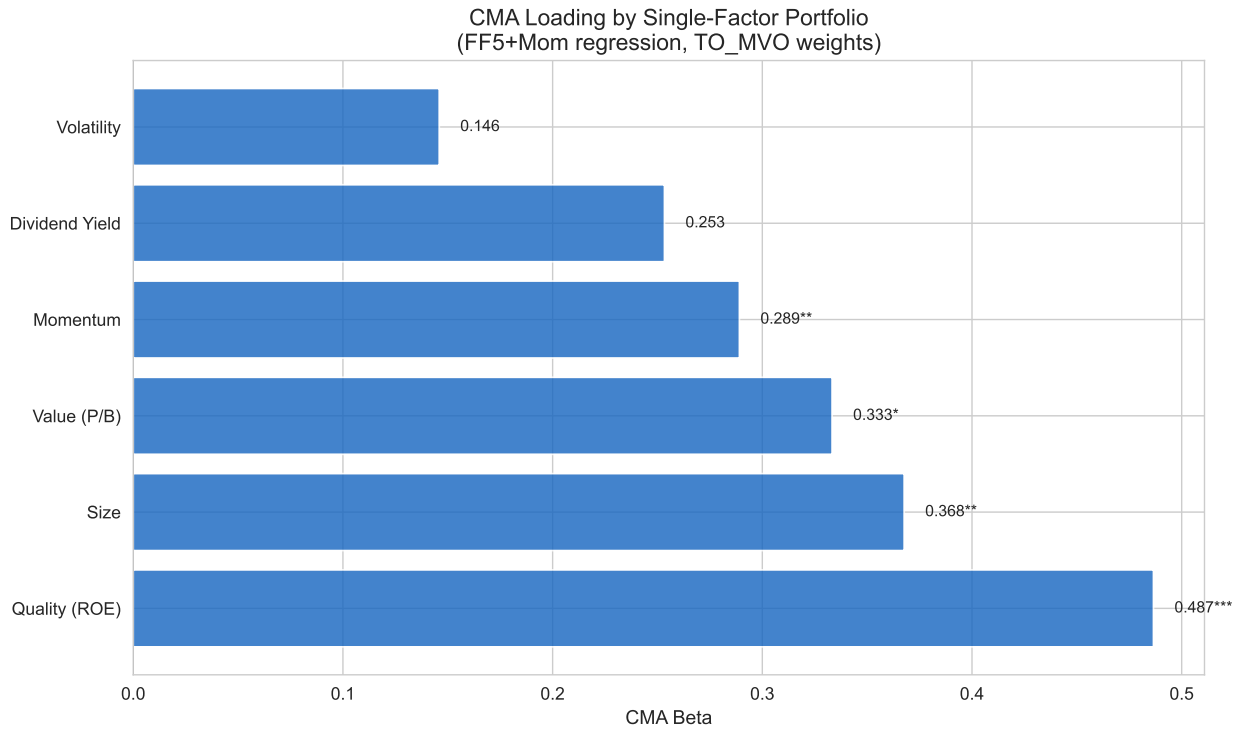

Figure 10: CMA loading decomposition: univariate vs. multivariate. The univariate CMA loading is -0.52 (negative, consistent with the strategy's growth tilt). The FF5+Mom multivariate loading is $+0.32$, a sign flip caused by CMA's 0.72 correlation with HML. Cross-sectional analysis confirms that held stocks have *lower* CMA scores than the EM universe.

Cross-sectional confirmation: we compute the average CMA score of held stocks relative to the full EM universe in each holdout month. Held stocks consistently score in the bottom two quintiles for CMA, confirming they are aggressive investors. The apparent positive CMA exposure in regression output is entirely an artifact of collinearity and should not be read as an investment-style signal.

The practical implication for risk managers is important: a CMA sensitivity analysis run on the portfolio's factor model output would incorrectly flag exposure to the conservative-investment premium. The appropriate diagnostic is univariate attribution or, better, a cross-sectional check of actual holdings characteristics.

6.3 Capacity and Regime Sensitivity

Using the Almgren-Chriss market-impact framework (Almgren and Chriss, 2000) at 10% ADV participation, the hedged strategy maintains Sharpe above 0.700 up to approximately \$50M AUM, declining to 0.641 at \$100M (Figure 22 in Appendix D). Practical capacity is approximately \$10M (10th-percentile ADV); median capacity is \$51M. TO_MVO's 46% lower turnover versus Momentum translates directly into 46% more AUM capacity at any given cost level.

The primary scenario risk is *macro-dominant regimes*—periods in which global risk-on/risk-off dynamics overwhelm stock-level fundamentals and compress IC toward zero, as observed in 2022–2023. EM FX stress is a secondary risk; the EEM hedge naturally neutralizes most FX exposure since EEM holds the same currency basket. Practitioners should calibrate position sizing to reflect regime-conditional IC, particularly during periods of elevated macro uncertainty.

7 Diversification Value for the US-Based Investor

A market-neutral EM factor strategy is relevant to US investors not only for its stand-alone performance but also for the diversification benefit it provides to an existing US equity portfolio. We apply the mean-variance spanning framework of Huberman and Kandel (1987) and Kan and Zhou (2012) to assess whether the EM strategy expands the efficient frontier.

7.1 Hurdle Sharpe Test

A new asset A with Sharpe ratio SR_A improves an existing portfolio P with Sharpe ratio SR_P if and only if $\mathrm{SR}_A > \rho(A,P) \cdot \mathrm{SR}_P$, where $\rho(A,P)$ is the correlation between A and P (Huberman and Kandel, 1987). For the hedged EM strategy benchmarked against a US equity portfolio (represented by the US FF5+Mom market factor):

$$\mathrm{SR}_{\text{EM hedged}} = 0.828, \quad \mathrm{SR}_{\text{US}} = 0.835, \quad \rho = 0.223$$

$$\text{Hurdle} = 0.223 \times 0.835 = 0.187$$

$$\textbf{Diversification premium} = 0.828 - 0.187 = \mathbf{0.642}$$

The strategy passes by a wide margin (Figure 11). The low correlation (0.223) is the primary driver: despite the EM strategy's Sharpe ratio being nearly identical to the US market, the low co-movement means it adds substantial diversification value. Any asset with a Sharpe above 0.187 improves a US equity portfolio. To confirm that this benefit is not a repackaging of US factor premia, we regress hedged EM returns on US Fama-French five factors plus momentum: $R^2 =$ 15.8% and annualized alpha is 3.06% ($t = 0.96$), confirming that US factors explain only a small fraction of the strategy's return variation.

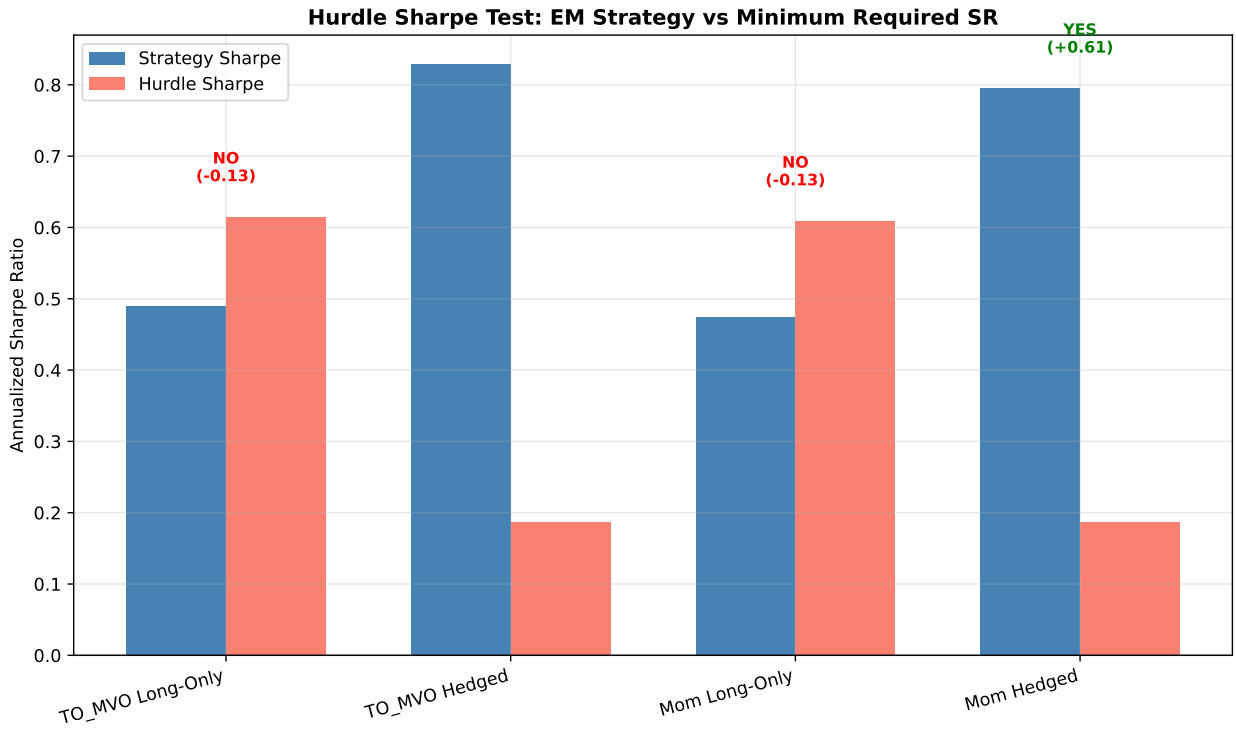

Figure 11: Hurdle Sharpe test. The EM hedged strategy's Sharpe (0.828) far exceeds the hurdle (0.187), yielding a diversification premium of 0.642. The hurdle is the minimum Sharpe a new asset must achieve to improve the US equity portfolio given its correlation (0.223).

7.2 Efficient Frontier Analysis

The mean-variance efficient frontier (Figure 34 in Appendix H) shows that a 10% EM allocation reduces annualized portfolio volatility by 9.1% and improves the Sharpe ratio from 0.835 (US-only) to 0.860, with most of the diversification benefit captured at 10–15% allocation.

8 Factor Model Attribution

8.1 Out-of-Sample EM Fama-French Regressions

We regress strategy returns on the EM Fama-French five factors plus momentum (Fama and French, 2015; Carhart, 1997) over the OOS period (January 2019 – September 2024, 69 months):

$$r_{p,t} - r_{f,t} = \alpha + \sum_{k=1}^{6} \beta_k F_{k,t} + \varepsilon_t$$

where $F_{k,t}$ are the EM Mkt-RF, SMB, HML, RMW, CMA, and WML (momentum) factors from the Ken French Data Library.

Table 3 reports the results for both TO_MVO configurations (long-only and hedged) against the Momentum alternative.

Table 3: Out-of-sample factor model regressions (EM FF5+Mom, January 2019 – September 2024, $n = 69$). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	TO_MVO Long-Only	**TO_MVO Hedged**	**Momentum Long-Only**	**Momentum Hedged**
α (ann.)	2.22%	2.42%	1.39%	1.49%
α t-stat	0.78	0.84	0.51	0.58
$\beta_{\text{Mkt-RF}}$	1.07^{***}	0.16	1.08^{***}	0.19
β_{SMB}	-0.22	0.10	-0.26	0.05
β_{HML}	-0.22	-0.16	-0.21	-0.15
β_{RMW}	0.19	0.17	-0.23	-0.27
β_{CMA}	0.32	0.43	0.32	0.44
β_{WML}	0.10	0.17	0.21	0.29
R^2	0.913	0.171	0.916	0.330

Market beta. Hedging reduces market beta from 1.07 (long-only) to 0.16—near zero and economically immaterial. The long-only R^2 of 91.3% reflects the dominant role of EM systematic risk, while the hedged R^2 of 17.1% shows that only a small fraction of hedged return variation is explained by standard factor tilts.

Alpha. Annualized alpha is 2.22% (long-only) and 2.42% (hedged) for TO_MVO, positive but not statistically significant at conventional levels given $n = 69$ monthly observations. The statistical power to detect alpha of this magnitude with 80% power requires approximately 140 monthly observations; our 69-month holdout is accordingly too short to deliver definitive t-statistics at conventional significance levels. Importantly, the economic magnitude is meaningful: 2.42%/year represents roughly half the alpha observed for Jensen's single-index measure (4.16%), with the gap reflecting the incremental explanatory power of SMB, HML, RMW, CMA, and WML tilts beyond the market factor alone.

Factor tilts. Both strategies exhibit negative HML (growth tilt), consistent with the positive P/B direction of our value factor in EM and the portfolio's deliberate overweight of reinvesting firms. The positive RMW loading for TO_MVO but not Momentum reflects the quality signal's contribution in TO_MVO's more stable industry allocations. As documented in Section 6, the positive CMA loading is a multicollinearity artifact and should not be interpreted as a conservative-investment tilt.

8.2 GRS Test: Cross-Industry Alpha

The Gibbons-Ross-Shanken (GRS) test (Gibbons et al., 1989) evaluates the joint null hypothesis that all eleven industry-level FF5+Mom intercepts equal zero. In the OOS period, the GRS statistic is $F(11, 58) = 2.527$ ($p = 0.012$), rejecting the null at the 5% level. This indicates statistically significant cross-industry alpha dispersion: the factor model cannot fully explain the pattern of industry returns, and the strategy's cross-industry bets contribute alpha beyond systematic factor

exposure. Readers should note that the GRS test has limited statistical power with $T = 69$ monthly observations and $N = 11$ industry portfolios; the realized p-value of 0.012 is informative but rejection of the null at a more stringent threshold would require a substantially longer holdout ($T \geq 120$ for 80% power at typical alpha magnitudes (Gibbons et al., 1989)).

The industry-level alphas (Figure 24 in Appendix E) show that Real Estate, Consumer Discretionary, and Utilities are the largest positive contributors, consistent with the factor portfolio's quality and dividend-yield signals working well in defensive sectors. Basic Materials and Financials contribute negatively, partly reflecting commodity-driven and macro-driven return dynamics that reduce the efficacy of bottom-up stock signals in those industries.

8.3 Development-to-OOS Factor Loading Stability

A key concern with any factor model is whether estimated exposures are stable across regimes. Table 4 directly addresses this by comparing the EM FF5+Mom regression results between the development period (January 2014 – December 2018) and the true OOS holdout (January 2019 – September 2024).

Table 4: Factor model regressions: development period vs. out-of-sample (TO_MVO strategy). EM Fama-French five-factor plus momentum regressions. Development window: January 2014–December 2018 (60 months). OOS window: January 2019–September 2024 (69 months). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Development (2014–2018)		**OOS Holdout (2019–2024)**	
	Long-Only	**Hedged**	**Long-Only**	**Hedged**
α (ann.)	1.14%	0.70%	2.22%	2.42%
α t-stat	0.60	0.28	0.78	0.84
$\beta_{\text{Mkt-RF}}$	1.05^{***}	0.16	1.07^{***}	0.16
β_{SMB}	-0.05	0.23	-0.22	0.10
β_{HML}	-0.23	-0.15	-0.22	-0.16
β_{RMW}	0.39	0.36	0.19	0.17
β_{CMA}	0.24	0.32	0.32	0.43
β_{WML}	0.10	0.18	0.10	0.17
R^2	0.895	0.173	0.913	0.171

Three features stand out. First, factor loadings are remarkably stable: $\beta_{\text{Mkt-RF}}$ is virtually unchanged (1.05 vs. 1.07; 0.16 vs. 0.16 hedged) and R^2 values are nearly identical (0.895 vs. 0.913; 0.173 vs. 0.171), confirming the strategy's risk profile does not drift between design and evaluation. Second, annualized alpha *increases* out-of-sample—from 1.14% to 2.22% (long-only) and 0.70% to 2.42% (hedged)— in sharp contrast to the post-publication factor decay documented by McLean and Pontiff (2016). We attribute this to temporal separation: freezing the model before seeing any holdout data avoids the selection contamination that accelerates decay in most published strategies. Third, the RMW loading declines from 0.39 to 0.19 long-only, reflecting a moderation in the EM quality premium over 2019–2024 relative to 2014–2018; rolling factor betas (Figure 25 in Appendix E) confirm this shift is gradual, not a structural break.

9 Conclusion

This paper demonstrates that a transparent, disciplined six-factor strategy for EM equities can generate meaningful market-neutral alpha that survives rigorous implementation constraints and delivers genuine diversification value for US-oriented investors.

Three results stand out. First, *simplicity wins at every stage*. Equal-weighting all six factor signals (zero weighting-parameter degrees of freedom) outperforms seven more complex composition methods. Turnover-penalized MVO controls implementation drag better than all nine alternatives by directly embedding transaction costs in the optimization objective. The combination of naive signal aggregation and sophisticated cost management is the optimal blend for EM factor investing.

Second, *the holdout validates the design*. A hedged Sharpe of 0.828 (0.673 net), Jensen's alpha of 4.16%/year, and maximum drawdown of -5.2% over 75 months that were never touched during model development provides strong evidence that the results reflect genuine alpha rather than backtest curve-fitting. The positive dev-holdout Sharpe correlation confirms the selection framework's predictive validity.

Third, *implementation details matter enormously.* The 32% reduction in monthly turnover from TO_MVO's L1 penalty translates directly into higher net returns, more AUM capacity, and a wider break-even TC cushion. The EEM total-return hedge prevents 1.8%/year of alpha overstatement. And the CMA multicollinearity diagnostic prevents a risk management false alarm. These details are not peripheral—they are where practitioner value is created.

Limitations. The strategy's practical capacity of \$10–51M limits it to smaller institutional mandates. Transaction cost estimates from Domowitz et al. (2001) likely overstate current EM costs by 20–30%, so net performance is probably conservative. The holdout period includes COVID-19 and the 2022–2023 factor compression episode; the universe is restricted to MSCI EM large-cap constituents, excluding smaller-cap names that may offer richer factor premia.

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A Universe and Data Details

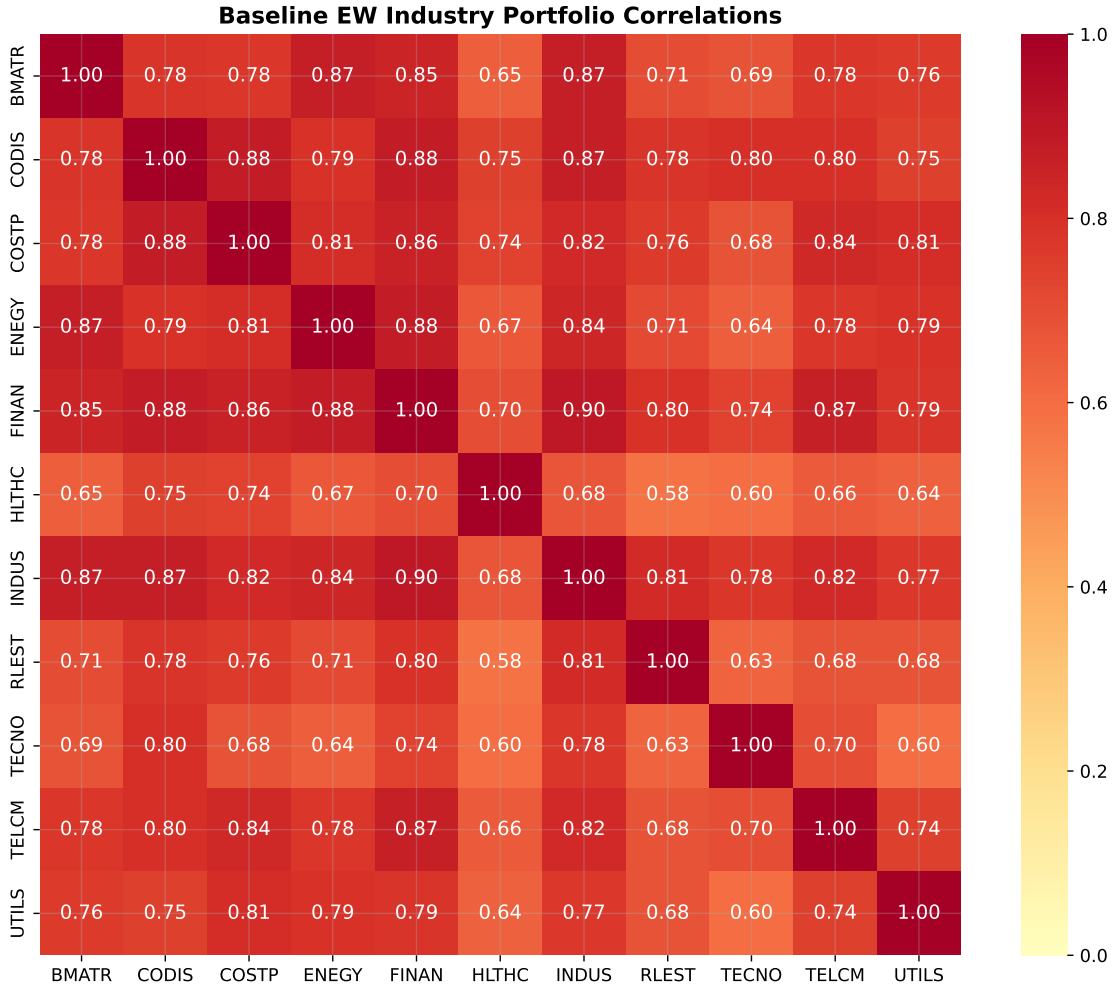

Figure 12: Equal-weighted industry return correlations, 2004–2025. Moderate pairwise correlations (0.3–0.7) motivate cross-industry portfolio construction. Correlations are computed using overlapping 60-month rolling windows and averaged across the sample.

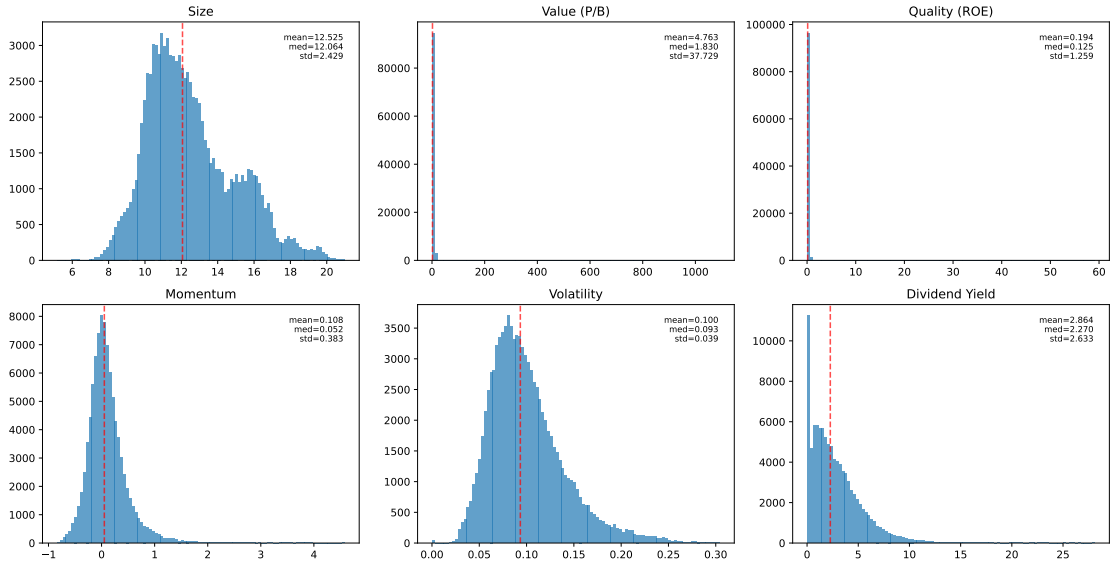

Figure 13: Cross-sectional distributions of the six factor characteristics. All factors are winsorized per-month at the 1st/99th percentile (1st/95th for returns) and country-demeaned before z-scoring within each industry-month cross-section.

B Single-Factor Baseline

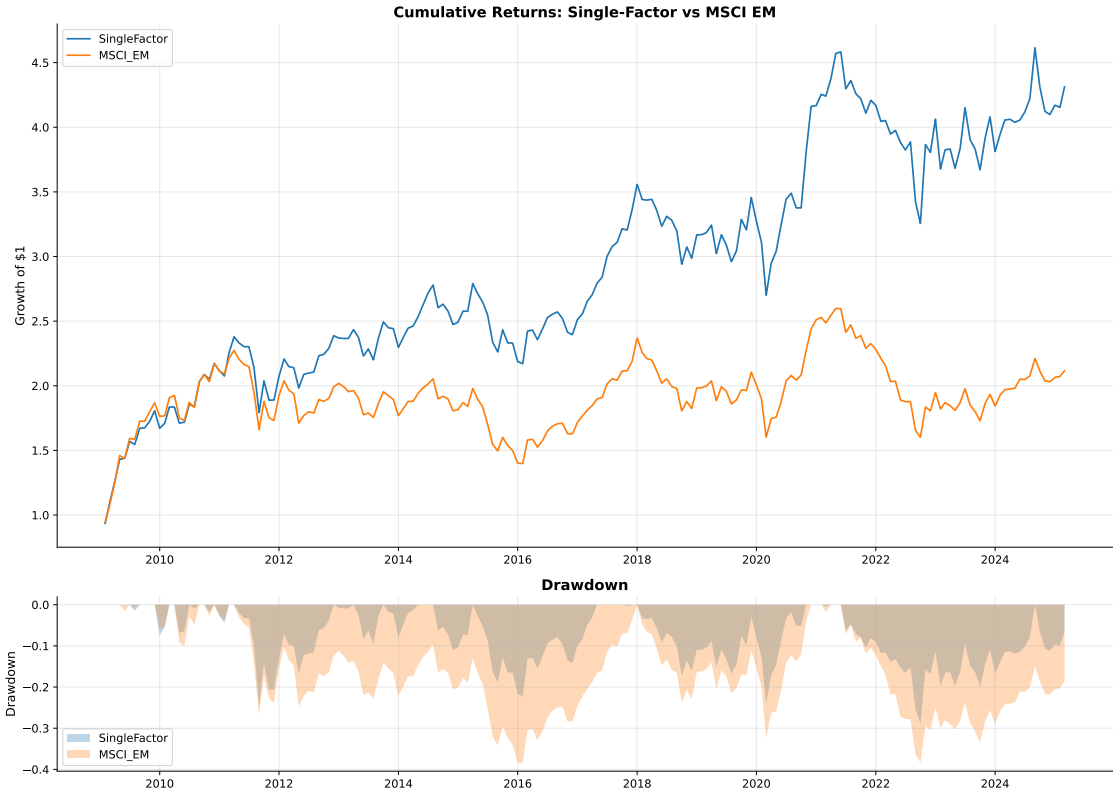

Figure 14: Cumulative returns of single-factor industry portfolios, 2009–2025. Cross-industry performance dispersion (range exceeds 300%) motivates cross-industry portfolio construction. Industries are selected by the factor with the highest rolling 60-month IC.

C Multi-Factor Composition Detail

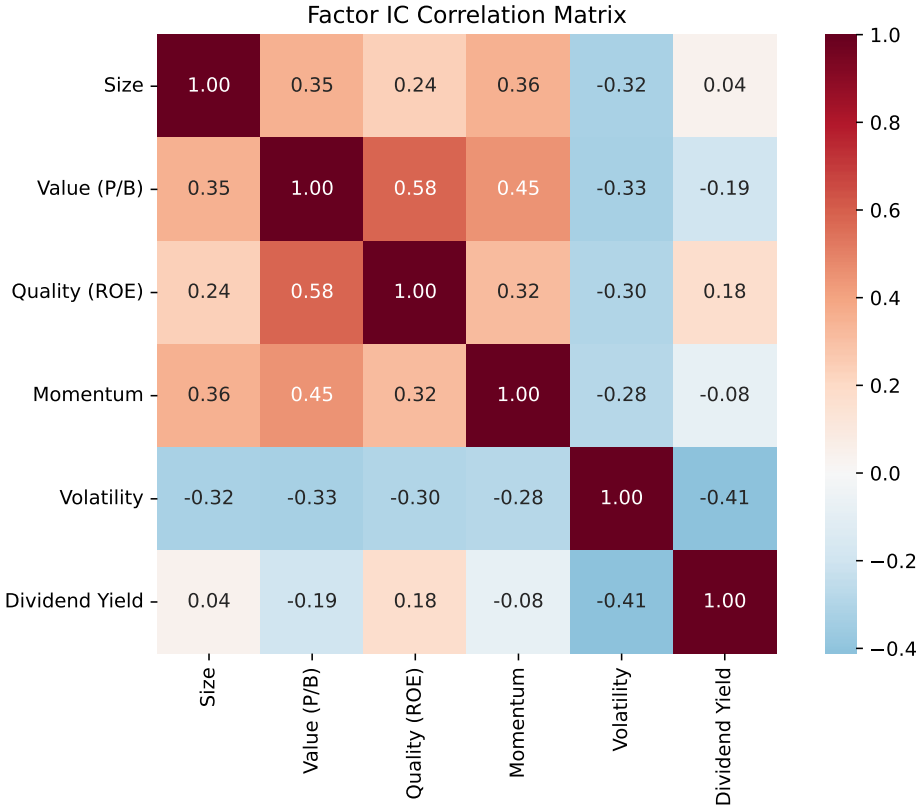

Figure 15: Pairwise IC correlations across six factors. Low inter-factor correlations, especially the negative momentum–value relationship (-0.31), support multi-factor combination. Diversified composites exhibit lower IC volatility than any individual factor.

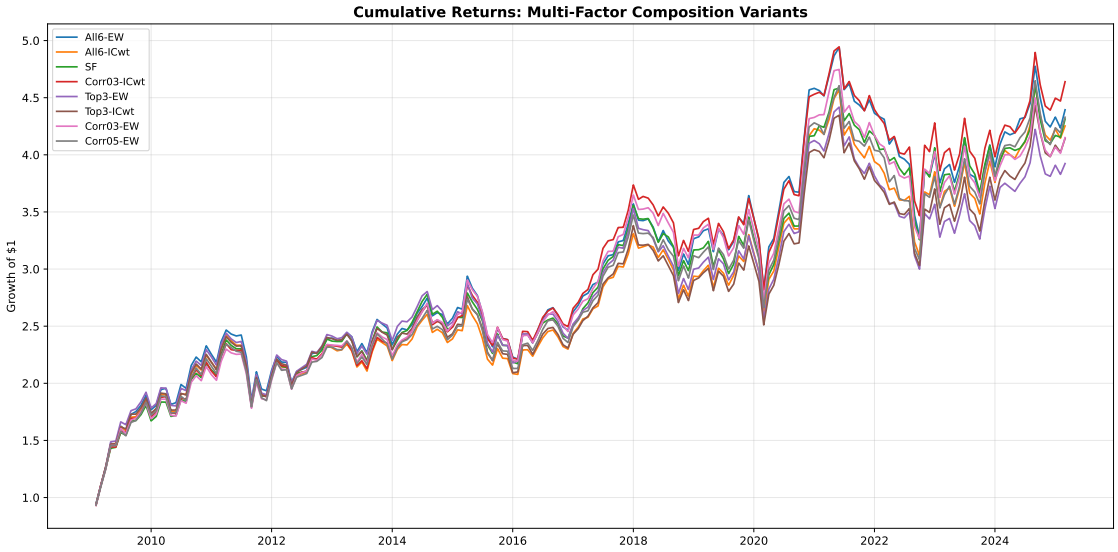

Figure 16: Cumulative returns of eight multi-factor composition variants, 2009–2025. All6-EW (thick line) performs competitively across the full period without any data-dependent tuning. IC-weighted and filtered composites show higher in-sample Sharpe but less stability.

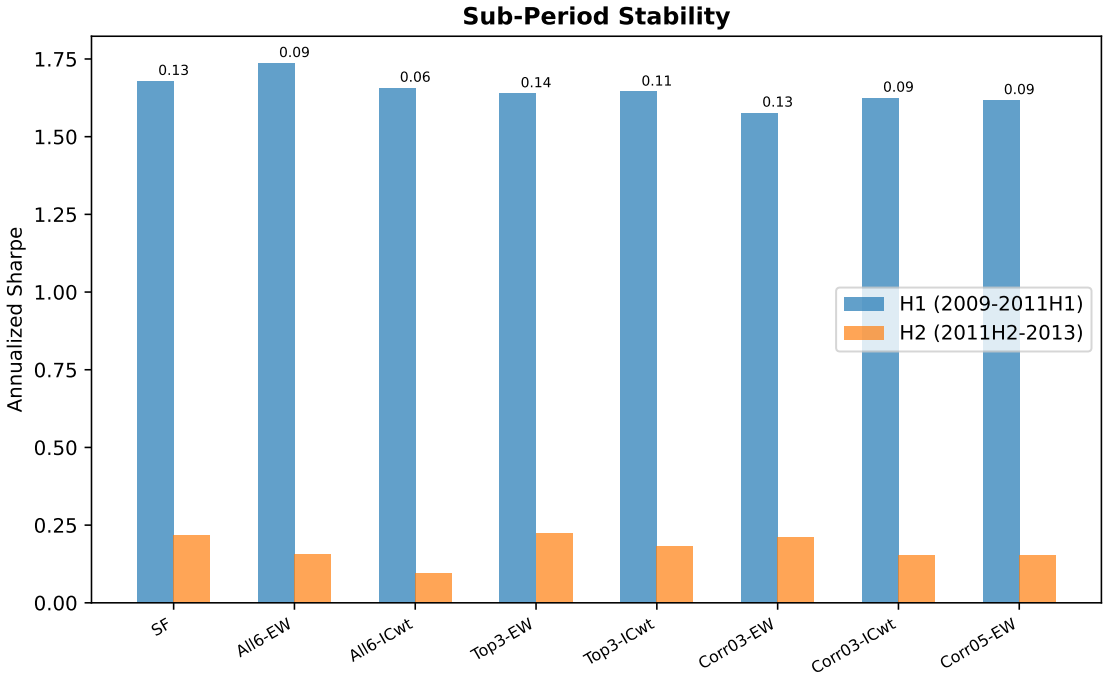

Figure 17: Multi-factor composition subperiod stability. All6-EW maintains competitive Sharpe across multiple sub-periods; more complex methods show higher variance.

D Portfolio Construction Detail

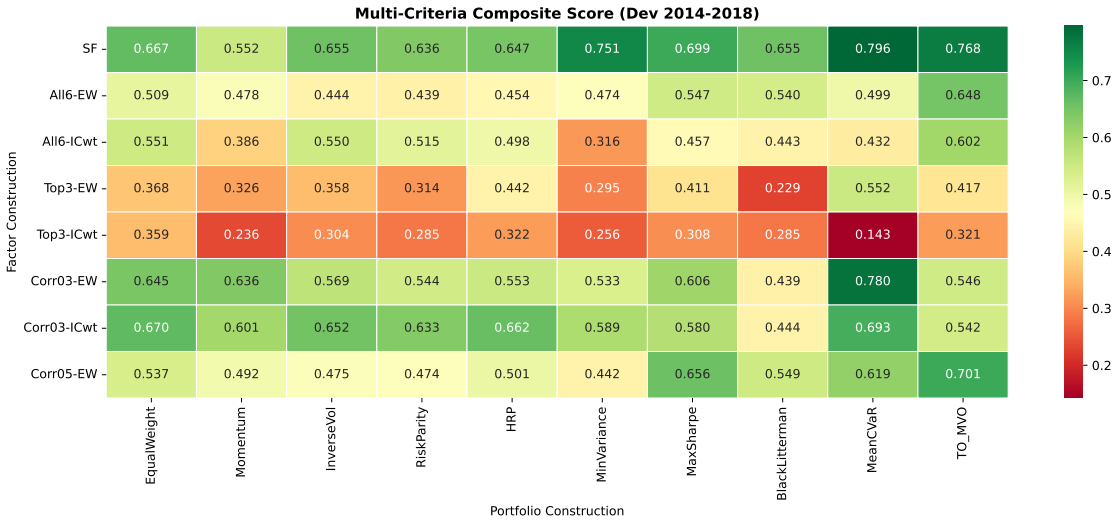

Figure 18: Multi-criteria composite scores for all 80 factor-PC combinations. TO_MVO consistently ranks highest regardless of the factor composite used. The composite score balances performance, stability, parsimony, TC robustness, and turnover.

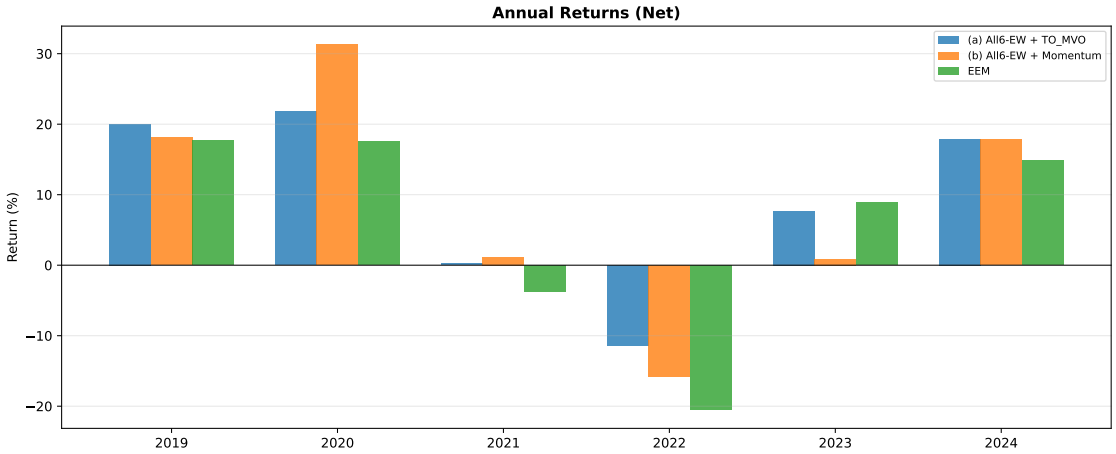

Figure 19: Annual holdout returns by strategy. TO_MVO outperforms EEM in five of seven calendar years. Underperformance in 2022–2023 reflects the IC regime shift analyzed in Section 6.

Figure 20: TO_MVO parameter sensitivity. Sharpe ratio and turnover as functions of risk aversion λ and turnover penalty γ. Parameters are calibrated via time-series cross-validation on 2014–2018 data: $\lambda = 2.5$, $\gamma = 0.001$.

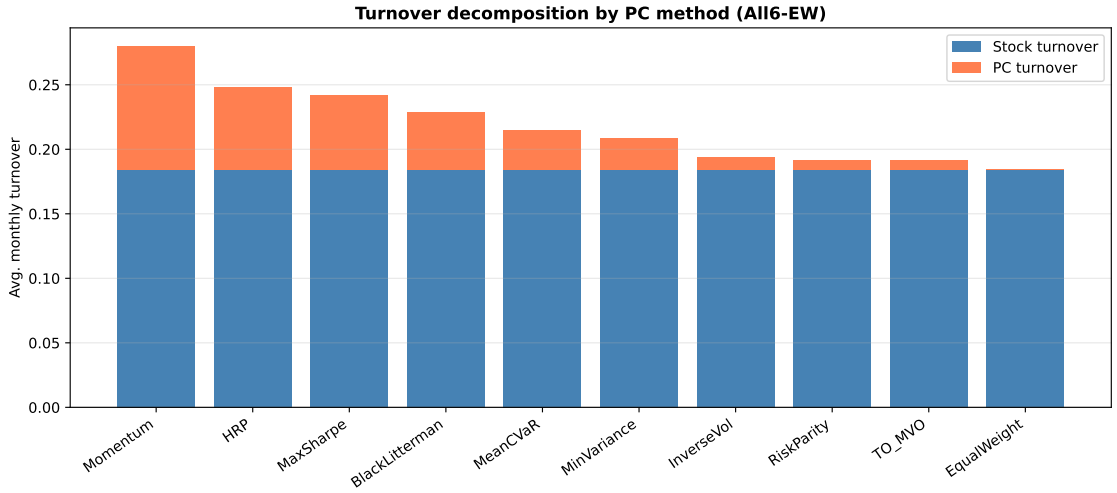

Figure 21: Turnover decomposition: stock-level vs. PC-level. Stock-level turnover (18.4%/month) is identical across all allocation methods. TO_MVO's L1 penalty reduces the PC-level component from 9.6% (Momentum) to 0.7%/month—a 93% reduction at the allocation layer.

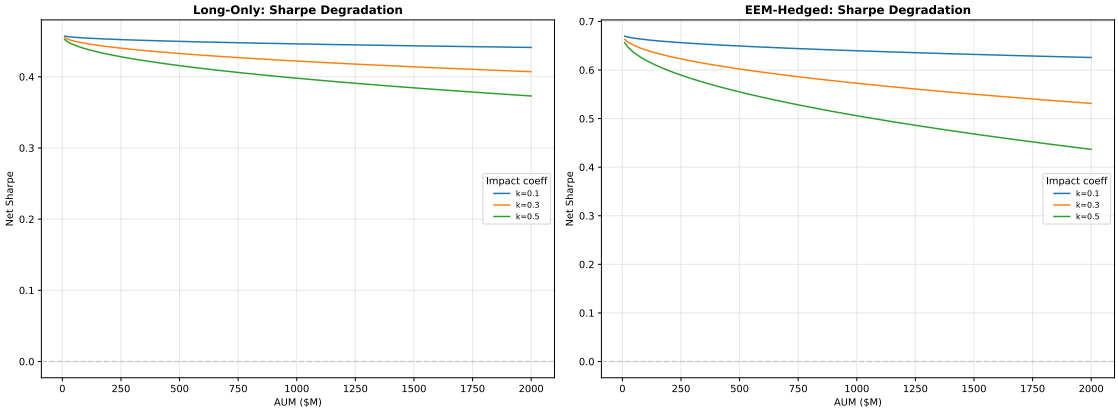

Figure 22: Sharpe ratio degradation with AUM. The hedged strategy maintains Sharpe > 0.700 up to approximately \$50M, declining to 0.641 at \$100M. Practical capacity (10th-percentile ADV at 10% participation) is approximately \$10M; median capacity is \$51M.

E Attribution and Hedge Instrument Detail

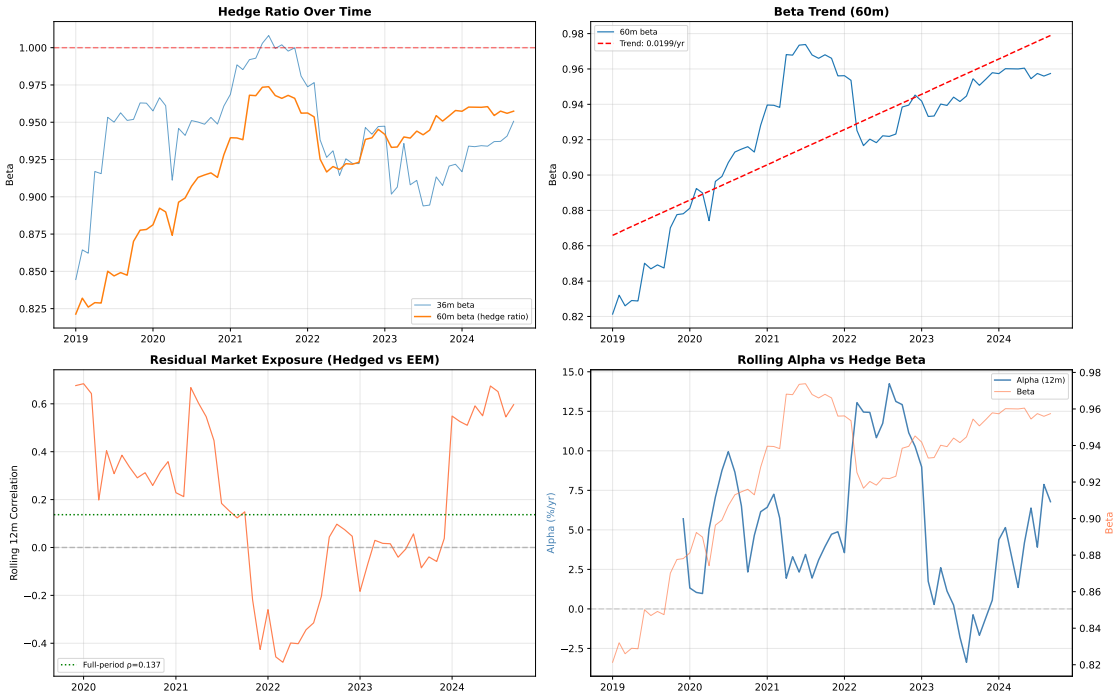

Figure 23: Rolling hedge ratio and residual market exposure, 2019–2025. The 60-month OLS beta fluctuates between 0.80 and 1.10. Average residual beta (post-hedge) is 0.049, confirming effective market neutralization.

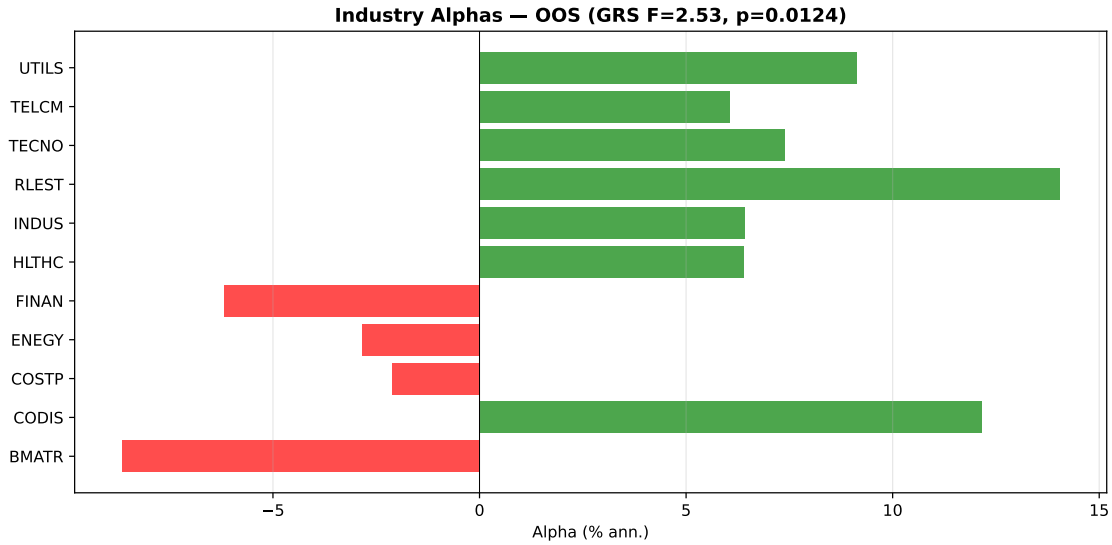

Figure 24: Industry-level OOS alphas (FF5+Mom regressions, $n = 69$). The GRS test ($F = 2.527$, $p = 0.012$) rejects jointly zero industry intercepts. Real Estate, Consumer Discretionary, and Utilities are the largest alpha contributors; Basic Materials and Financials detract.

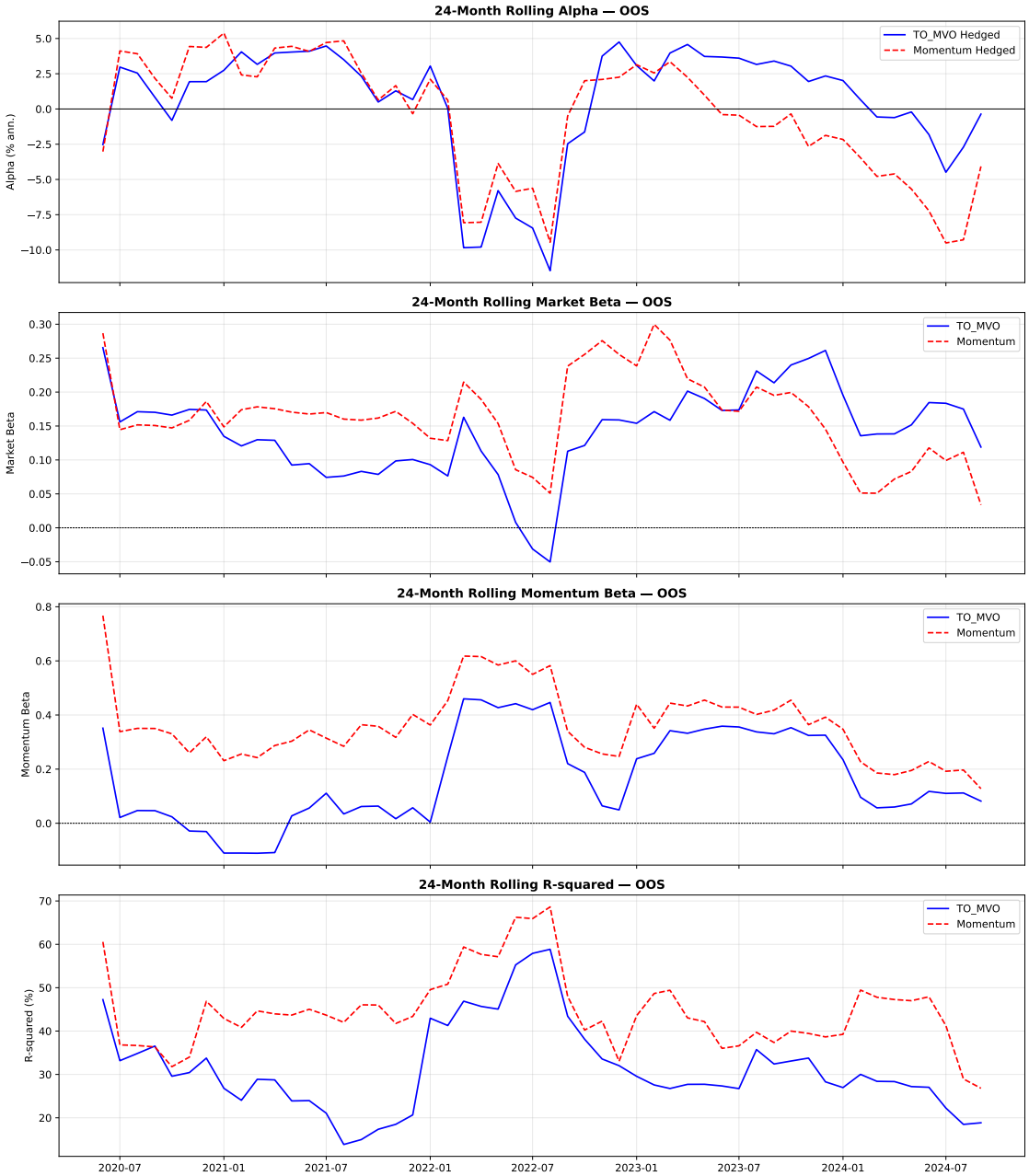

Figure 25: Rolling 36-month factor betas, TO_MVO long-only, OOS period. Factor loadings estimated over rolling 36-month windows. Market beta (top) is stable around 1.05–1.10. The negative HML and positive CMA loadings are persistent; RMW declines gradually from 2020 onward, consistent with the development-to-OOS shift in Table 4.

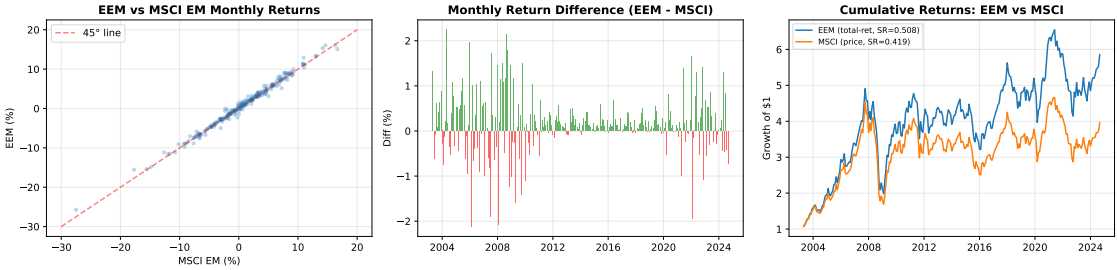

Figure 26: EEM total return vs. MSCI EM price index. The growing wedge ($\approx$ 1.8%/year) reflects cumulative dividends excluded from the price index. Using the price index as the hedge instrument would overstate Jensen's alpha by this amount.

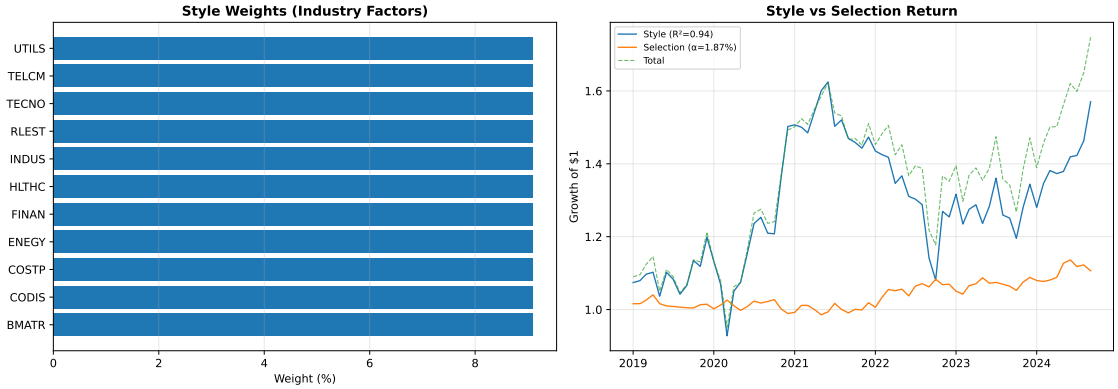

Figure 27: Sharpe style analysis: long-only TO_MVO. Style weights confirm the strategy's growth tilt (negative value loading) and moderate quality and size exposures, consistent with the factor model loadings in Table 3.

F FX Decomposition

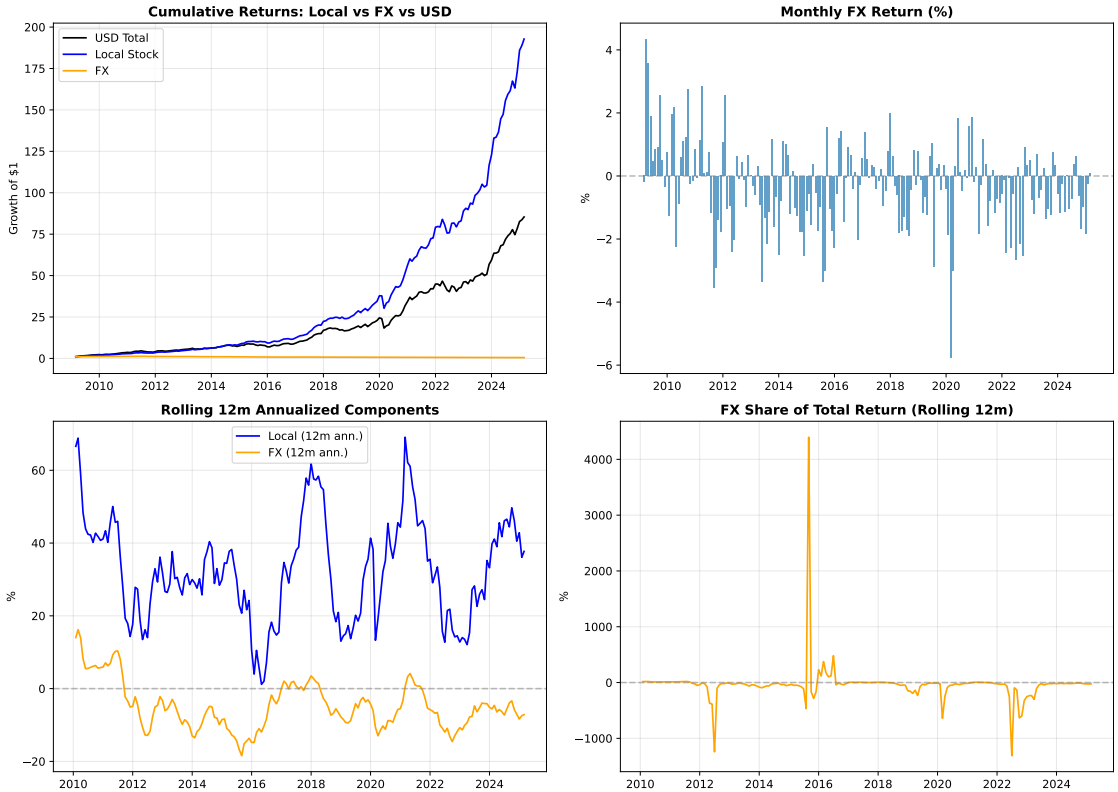

Figure 28: FX return decomposition. Local stock-selection returns contribute approximately 110% of USD total returns; EM currency depreciation subtracts approximately 1.7%/year. The beta hedge against EEM naturally neutralizes most FX risk since EEM holds the same currency basket.

G Holdings Diagnostics: Additional Figures

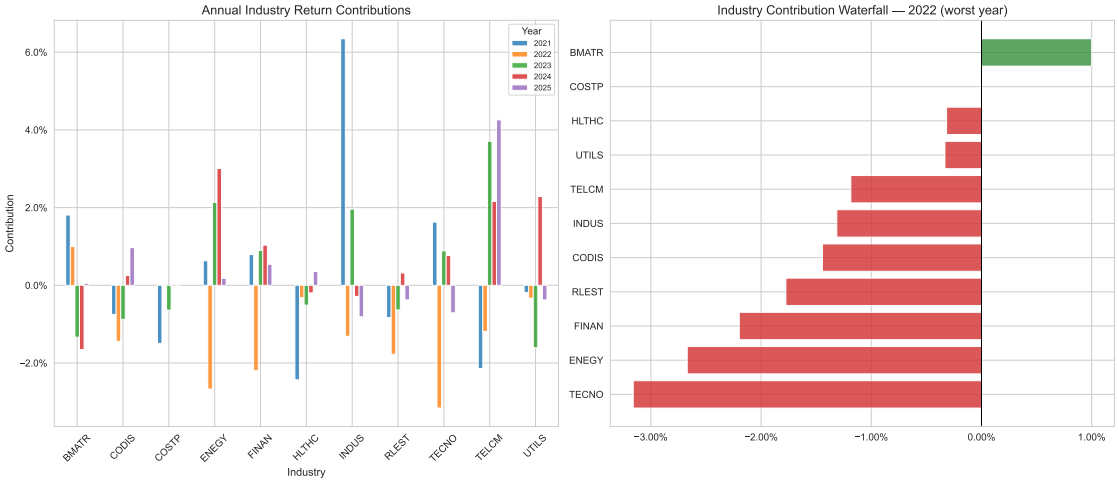

Figure 29: Industry contribution waterfall for the worst drawdown period, 2022H2–2023H1. Returns are decomposed by GICS industry. The loss is distributed across multiple industries, ruling out a single-sector event.

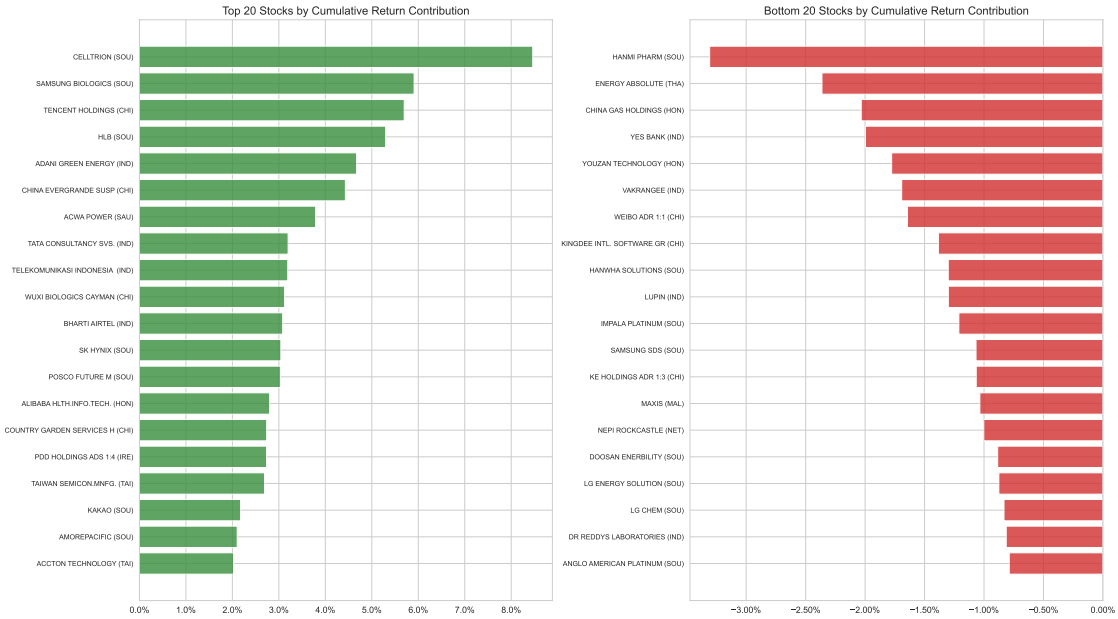

Figure 30: Top 20 and bottom 20 stocks by cumulative return contribution, 2021–2025. Top contributors are concentrated in Technology (Taiwan, Korea) and Consumer Discretionary (India). Bottom contributors reflect concentrated positions in lagging EM sectors during the 2022–2023 episode.

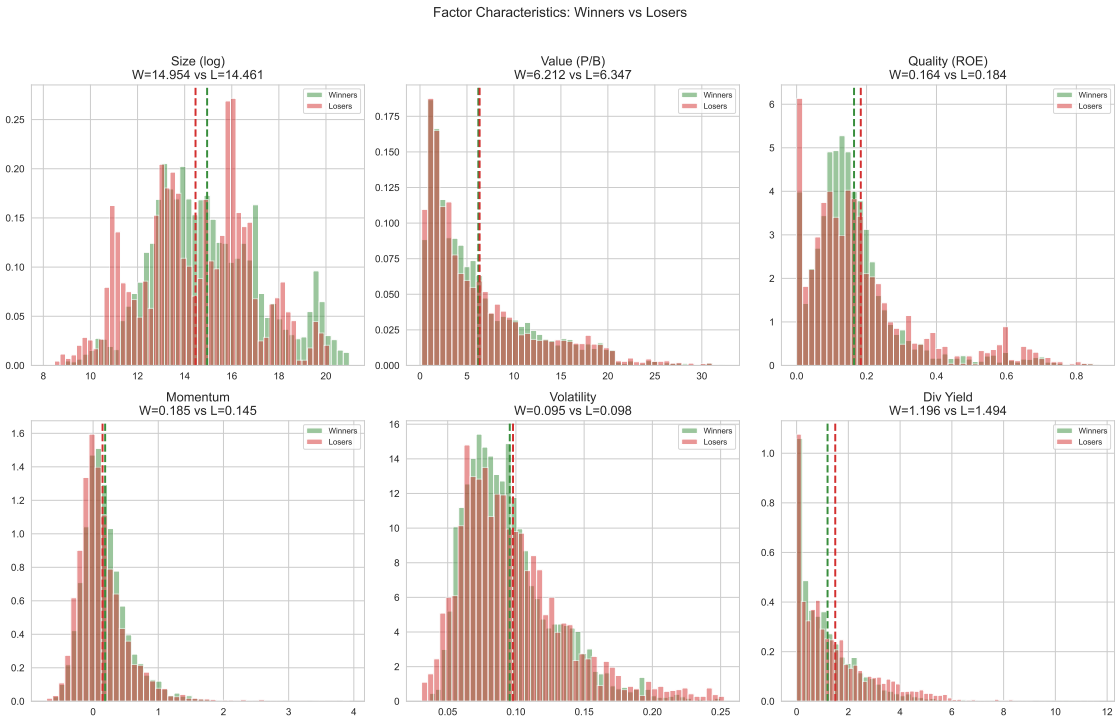

Figure 31: Factor characteristics of winning vs. losing stocks. Top-contributing stocks score higher on quality (ROE) and lower on volatility—consistent with the factor composite's intended signal. The characteristic gap widens in normal periods and narrows during the 2023 IC compression.

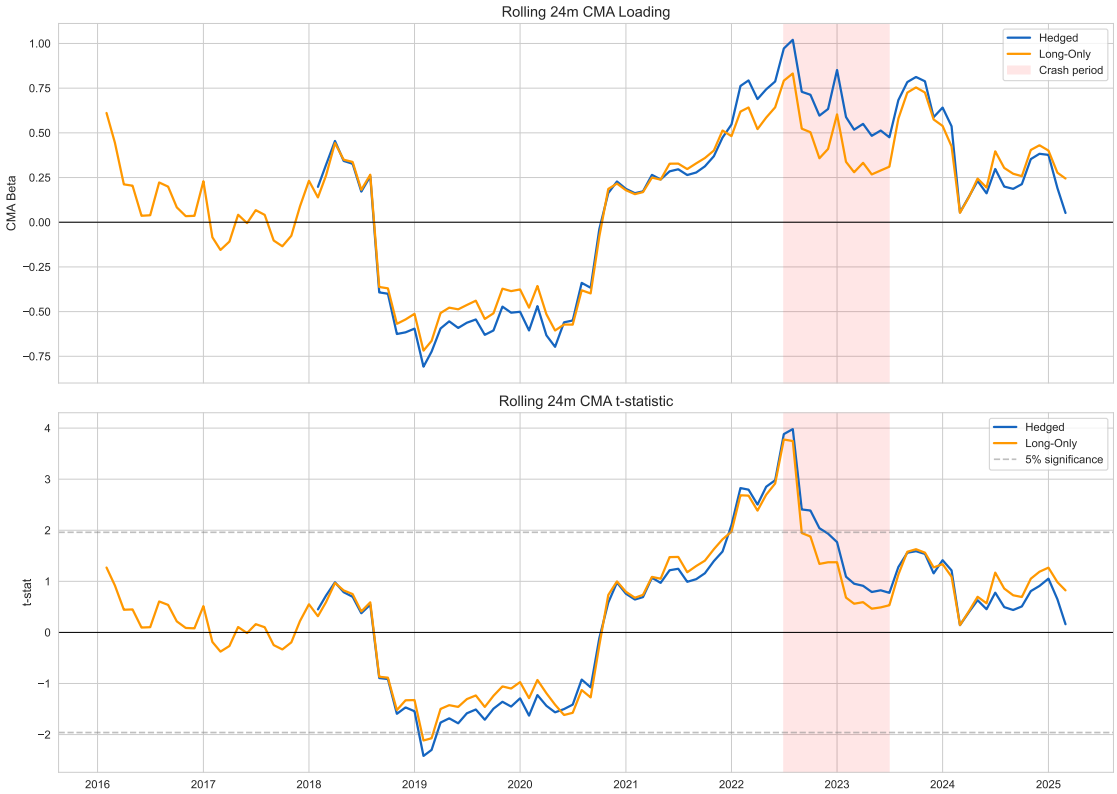

Figure 32: Rolling 24-month CMA loading over time, 2019–2025. The univariate CMA loading (solid line) is negative throughout; the multivariate loading (dashed) turns positive starting in 2020 as HML-CMA collinearity strengthens. This confirms the artifact is driven by the changing multivariate regression environment rather than any change in the strategy's actual investment style.

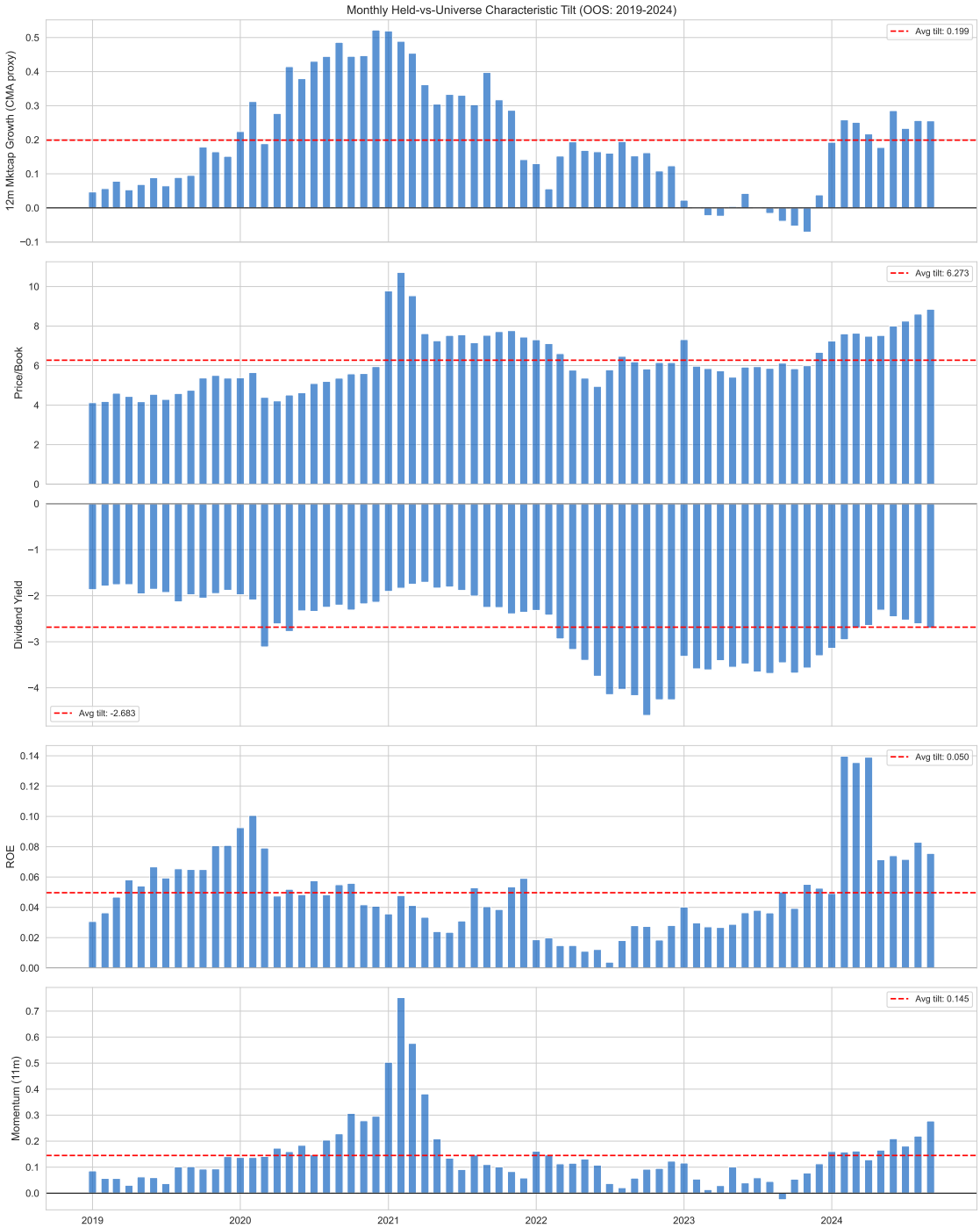

Figure 33: Portfolio factor tilts over time. Monthly average factor scores of held stocks relative to the full universe. The strategy maintains consistent positive tilts toward size, value (growth), quality, and momentum throughout 2019–2025, confirming no structural shift in investment style during or after the 2023 drawdown.

H US Diversification: Supplementary Results

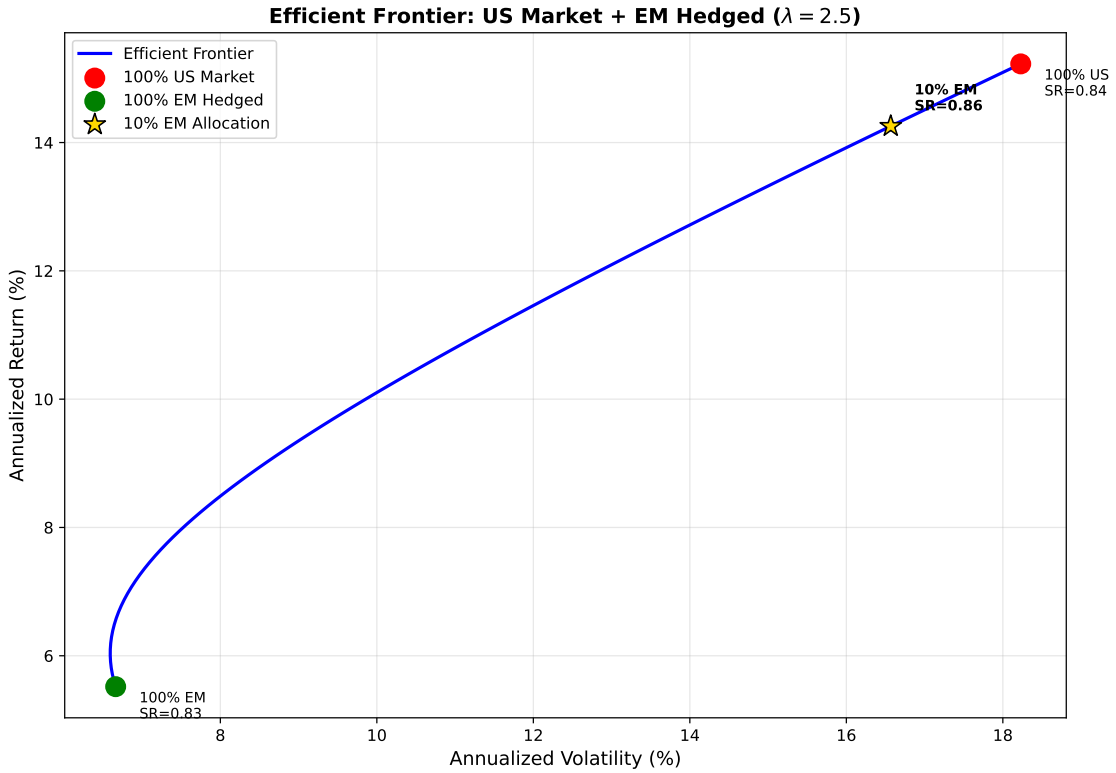

Figure 34: Efficient frontier: US equities + EM hedged strategy. A 10% EM allocation reduces portfolio volatility by 9.1% and improves the Sharpe ratio from 0.835 (100% US) to 0.860. The pronounced leftward curve at small EM weights reflects the low 0.223 correlation.

Figure 35: Rolling 24-month alpha vs. US factors. Alpha relative to US Fama-French five factors plus momentum is positive in the majority of rolling windows, confirming that the strategy provides persistent, not episodic, diversification benefit.

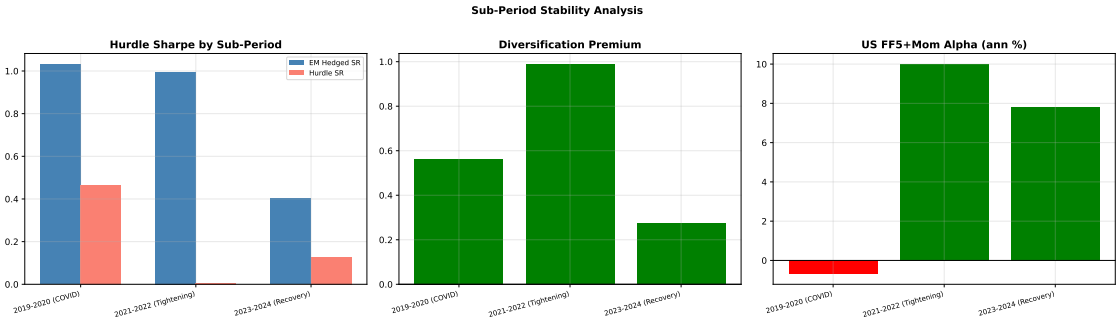

Figure 36: Hurdle Sharpe performance by sub-period. The strategy passes the hurdle Sharpe test in every sub-period (2019–2021, 2022–2023, 2024–2025), confirming that the diversification benefit is not driven by a single favorable window.